

Multi-Unit Search: An Experimental Approach^{*}

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September 11, 2026

Abstract

We experimentally explore sellers' dynamic pricing strategies in a multi-unit search environment with stochastic buyer arrivals. Building on Carrasco and Smith (2017), our laboratory design examines how reservation prices adapt to inventory levels and demand distributions, as well as to deadlines. Despite theoretical predictions suggesting that higher inventory levels and imminent deadlines reduce reservation prices, our findings reveal negligible effects of both factors on pricing behavior. Finite look-ahead does not improve fit, while simple subject-specific pricing rules predict behavior substantially better out of sample than the rational benchmark. Our experimental results reveal previously unexplored dynamics in multi-unit search environments, underscoring the need for further empirical research to bridge the gap between theory and observed seller behavior.

JEL classification: C70, C91, D25, D83

Keywords: Multi-unit search; reservation prices; inventory level; deadlines; supply curves

^{*}This study received ethical approval from UQ's BEL LNR Committee (2023/HE002373) and was pre-registered in the AEA RCT registry (AEARCTR-0014339). Jose A. Carrasco thanks and acknowledges financial support from FONDECYT #1221816 and FONDECYT #1251698.

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1 Introduction

Search models provide elegant and powerful theoretical frameworks for analyzing the incentives that shape equilibria in decentralized environments, especially in settings with transaction frictions in labor, monetary, and financial markets. To obtain tractable equilibrium predictions, however, classical search models typically restrict attention to single-unit environments. Carrasco and Smith (2017) relax this restriction by developing a framework in which sellers hold multiple indivisible units and face stochastic trading opportunities. A central implication of their analysis is that optimal reservation prices depend on the seller’s inventory and, in particular, decline as the number of units held by the seller increases.

The framework proposed by Carrasco and Smith (2017) faces significant practical challenges in empirical validation, primarily due to the scarcity of datasets that capture multi-unit trading dynamics. Datasets from digital platforms such as eBay or Ticketmaster could be valuable for studying multi-unit trade, but access may be restricted by proprietary-data, privacy, and confidentiality constraints (Einav and Levin, 2014). Moreover, the applicability of these insights extends beyond online platforms to other industries, such as airlines selling multiple tickets for the same flight or hotels managing room availability. In these settings, multi-unit pricing and selling strategies are essential. However, the limitations of available datasets are even more pronounced in these industries, where competitors could infer opponents’ strategies from dynamic pricing behavior. Given these empirical challenges, experimental approaches offer a valuable alternative for studying these theoretical predictions.

In this paper, we test behavioral predictions in a setting where the seller holds multiple units and faces the periodic arrival of trade opportunities, as in Carrasco and Smith (2017). Specifically, we investigate how reservation prices evolve over time, adjust with changes in the seller’s inventory, and respond to the presence of deadlines. Our goal is to advance the emerging theoretical literature on multi-unit search models by experimentally evaluating their main predictions in the laboratory, providing empirical evidence on the dynamics of multi-unit pricing and seller optimal pricing behavior.

Experimental design We develop a controlled environment to examine how individuals make dynamic selling decisions when managing multiple indivisible units of a homogeneous

good. Demand for the goods is experimentally controlled and stochastic. Participants act as sellers, determining reservation prices for each quantity that they might sell. Our experiment was implemented online through Prolific, and our main analyses were pre-registered in the American Economic Association’s RCT Registry (AEARCTR-0014339).

We adopt a mixed between- and within-subjects design to study the effects of changes in demand and inventory, as well as the consequences of implementing hard deadlines. The within-subject treatment varied the inventory level and demand distributions that subjects faced in their search problems. The ordering of these parameter sets was randomized at the subject level. The between-subject treatment implemented a deadline: if subjects do not sell all of their inventory by the (fixed and known) deadline, all the remaining inventory loses its value.

During each trading stage, participants set minimum acceptable prices (i.e., reservation prices) up to either their maximum inventory or the maximum possible demand in the stage, whichever is lower. Prices and payments are denominated in terms of lottery tickets in order to remove the potential for risk aversion to affect reservation prices.¹ We simulate discounting by imposing a termination chance after each stage. The subjects’ search problem concludes when all units are sold, the stochastic continuation process ends, or a deadline is reached.

Findings The main outcome variables of interest are the reservation prices that subjects report for each decision problem. Specifically, we study: (i) the effect of inventory levels on reservation prices, (ii) the effect of deadlines on reservation prices, and (iii) the determinants of deviations between empirical and theoretical reservation prices.

Based on eight configurations with varying inventory levels and demand distributions presented in random orders to subjects, we do not find support for the main theoretically predicted effects of changes in inventory and deadline. While higher inventory levels are expected to lower reservation prices, the empirical effect is negligible and statistically insignificant, suggesting no detectable impact in this setting. Deadlines also have no discernible effect on reservation prices, contradicting the prediction that time pressure should lead sellers to lower their prices.

Finally, we find that deviations from theoretical predictions are positively associated with

¹Previous work has accounted for risk aversion when comparing behavior to theory (see, for instance, Schunk (2009), Schunk and Winter (2009), and Horváth (2023)). To the best of our knowledge, this paper is the first to use payments denominated in probability in order to *eliminate* the effect of risk aversion (albeit under the assumption of expected utility maximization).

inventory and unit number, while higher average demand reduces deviations. These results suggest that deviations mainly reflect systematic changes in theoretical predictions rather than significant behavioral differences across treatments. Notably, we find that absolute deviations grow modestly over time, implying that experience does not lead to convergence toward theoretical predictions and may instead be associated with fatigue or disengagement.

Related literature Our paper relates most directly to the theoretical literature on multi-unit search. While classical search models typically consider single-unit environments, Carrasco and Smith (2017) develop a framework in which sellers manage inventories of multiple indivisible units and characterize optimal reservation-price policies under stochastic trading opportunities. Subsequent work extends this framework to costly search (Carrasco and Harrison, 2023) and buyers with downward-sloping demand (Carrasco et al., 2024). We contribute to this literature by providing, to our knowledge, the first experimental test of the behavioral predictions of multi-unit search theory. In particular, we examine how reservation prices respond to inventory and demand restrictions, and how these responses change in the presence of a deadline.

Our experiment also builds on the experimental literature on optimal search. Early laboratory studies provide mixed evidence on the accuracy of theoretical search models. Schotter and Braunstein (1981) and Braunstein and Schotter (1982) report treatment effects that are broadly consistent with the theoretical comparative-static predictions. Cox and Oaxaca (1989) find close agreement between theoretical predictions and their experimental results, while Cox and Oaxaca (1992) show that tests using elicited reservation wages reject the risk-neutral model but not the more general concave-utility model. Sonnemans (1998) finds that subjects frequently terminate search earlier than predicted, while Hey (1982) documents the use of simple search and stopping rules in laboratory search tasks. Our single-unit environment provides a natural benchmark against this literature; indeed, our finding that reservation prices are below the theoretical benchmark echoes the early-search behavior documented by Sonnemans (1998). Our main analysis moves beyond this literature by studying environments in which sellers hold multiple units and must account for both current inventory and future trading opportunities.

A related experimental literature studies why decisions depart from optimal dynamic policies. Carbone and Hey (2004) find that subjects over-react to changes in their employment

status, although the theory’s comparative-static predictions are largely confirmed. Carbone (2005) documents substantial heterogeneity in decision-making, with some subjects behaving close to the theoretical optimum and others relying on simple heuristic rules. Similarly, Caplin et al. (2011) provide evidence consistent with satisficing in search decisions. More broadly, Enke and Graeber (2023) show that cognitive uncertainty about the optimal action can attenuate responses to changes in objective parameters. These mechanisms are particularly relevant in our setting because inventory and deadlines affect reservation prices through continuation values. Responding optimally therefore requires sellers to assess how changes in the current state alter the value of future trading opportunities.

Our analysis also connects to experimental work on inventory management. Schweitzer and Cachon (2000) document systematic departures from expected-profit-maximizing inventory choices, while Bolton and Katok (2008) show that experience reduces, but does not eliminate, such departures, and Lurie and Swaminathan (2009) study the role of feedback in inventory decisions. These papers study inventory choice rather than search. In our environment, inventory is instead a state variable of the dynamic selling problem: the number of units remaining determines their marginal value and, consequently, the seller’s optimal reservation prices.

Finally, our deadline treatment relates to experimental work on finite-horizon search. Cox and Oaxaca (1989) compare behavior across search horizons, while Baumann et al. (2023) examine adaptation to horizon length in sequential search and Marcu and Noussair (2024) study how the search horizon interacts with the ability to retain previously observed offers. We complement this literature by introducing a deadline into a multi-unit search environment, where the value of waiting depends jointly on the remaining horizon and the seller’s inventory.

Our experiment then studies the distinctive intertemporal trade-offs that arise when sellers manage multiple units over time and provides evidence on whether reservation prices respond to demand, inventory, and the search horizon as predicted by multi-unit search theory.

2 Theoretical predictions and hypotheses

In this section, we present the theoretical framework used to derive our main results, which are the foundation of our experimental hypotheses. We outline the minimal environment able to

capture key features of multi-unit search models, including discounting, demand distributions, inventory levels, and deadlines. The stationary characterization of value functions follows directly from Carrasco and Smith (2017) and so do most of our theoretical predictions. We adapt their framework to our binary-lottery payment mechanism without a deadline (Section A.1) and with a finite horizon (Section A.2), and compute the theoretical predictions for the configurations used in the experiment.

2.1 The environment

We consider a discrete time framework with a potential deadline T , where each trading stage is labeled as $t = 1, 2, 3, \dots, T$. In every stage $t < T$, there is a constant probability $\beta < 1$ that time will continue into the next stage. Naturally, at the deadline T no further continuation is possible, effectively terminating any decision-making process. At the beginning of each stage t , a seller holds an inventory of goods $n_t \in \{1, 2, 3, \dots\}$ consisting of homogeneous and indivisible items (e.g., flight tickets, hotel rooms) and has earnings of $v_t \in \mathbb{R}_+$. Initial earnings are $v_1 = 0$.²

Trade opportunities arise sequentially, and the seller receives only one offer per stage, reflecting the uncertainty inherent in any trading process. In each stage t , the seller is presented with a random offer (p_t, q_t) , where p_t represents the per-unit price, and q_t denotes the demand size.³ Upon receiving an offer (p_t, q_t) , the seller may choose to partially accept it by selling any quantity $y_t \in \{0, 1, 2, \dots, \min(n_t, q_t)\}$. After selling y_t units at price p_t , the seller's remaining inventory becomes $n_{t+1} = n_t - y_t$, and cumulative earnings update to $v_{t+1} = v_t + p_t y_t$.

Once a sale is made, the seller proceeds to the next stage without the option to revise previous decisions. This irreversibility highlights the importance of strategic decision-making at each stage. The trading process ends when the seller has no units left to sell or when time reaches the deadline T , introducing a natural constraint on trading opportunities and emphasizing the interplay between time and decision-making.

Price offers and demand sizes are drawn independently each stage. Price offers p_t are drawn

²In our experiment, both price offers and earnings are denominated in lottery tickets, where each ticket represents a 1% chance of winning a \$10 prize. This design ensures that expected utility maximizing choices align with expected value maximization. This is consistent with our theoretical framework, which assumes expected value maximizing behavior.

³We follow the standard probability convention in which uppercase letters (e.g., P, Q) denote random variables, and their lowercase counterparts (e.g., p, q) represent specific realizations.

from the discrete uniform distribution over integers between 1 and $\bar{p}$. Demand sizes take discrete values $\ell = 1, 2, 3, \dots, m$ with associated probabilities $\alpha_\ell \geq 0$, satisfying $\alpha_1 + \alpha_2 + \dots + \alpha_m = 1$. This structure introduces variability into the model, providing a framework to examine how demand distributions shape seller's decisions.

2.2 Reservation Prices

In general, let $\mathcal{W}_t(n)$ denote the value of holding n units after trading in stage t . For $t < T$, as offers are idd across stages, this value function satisfies the following recursive relation:

$$\mathcal{W}_t(n) = \beta \mathbb{E} \left(\max_{y \in \{0, 1, \dots, \min(n, Q)\}} [Py + \mathcal{W}_{t+1}(n - y)] \right),$$

with $\mathcal{W}_T(n) = 0$, where β is the discount factor, Q is the random demand, and P the random price. Furthermore, as we show in Proposition A.1 in the Appendix for the case without a deadline, and in Proposition A.2 for the case with a deadline, the value $\mathcal{W}$ is concave in n , which means reservation prices are given by the slope of the value function.

Without a deadline, the problem is stationary, and we refer to the value function in this case as $\mathcal{W}(n)$. Due to the concavity of the value function, the reservation price for selling i units with an inventory of n is

$$\mathcal{R}_{i,n} = \mathcal{W}(n - i + 1) - \mathcal{W}(n - i).$$

Figure 1 graphically shows $\mathcal{W}(n)$ and $\mathcal{R}_{i,n}$ for some of the parameters we use in the experiment.

Our first hypothesis describes optimal reservation prices when demand is not restricted.

Hypothesis 1 (Reservation prices without demand restrictions) *All reservation prices without demand restrictions are equal.*

Hypothesis 1 follows from Corollary 1 in the Appendix. It shows that when the seller has multiple units in inventory but faces no constraints on buyers' demands, the problem reduces mathematically to the classical single-unit search model. As a result, the optimal reservation price coincides with that of the one-unit setting. This occurs because, without

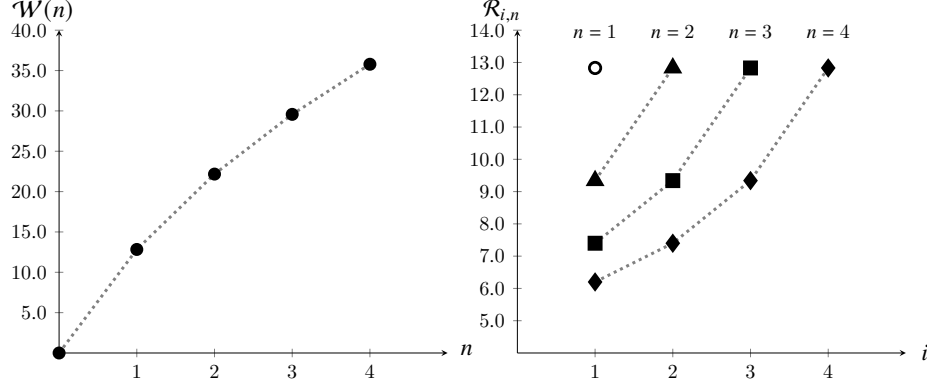


Figure 1: **Values and Reservation Prices without Deadline.** We assume that demands are either 1 or 4, occurring with probabilities $\alpha_1 = 0.9$ and $\alpha_4 = 0.1$, respectively. The continuation probability is $\beta = 0.8$, and prices are discrete and uniformly distributed over the interval $[1, 25]$. The left panel displays the value function $\mathcal{W}(n)$, and the right panel shows the corresponding reservation prices, which vary with the seller's inventory.

demand restrictions, the intertemporal trade-offs inherent in inventory management vanish, leading to pricing behavior identical to the simpler case. Corollary 4 in the Appendix further extends this equivalence to environments with a deadline.

Hypothesis 2 (Sell-all price) *The reservation price to sell all units is strictly positive and equal to the single-unit reservation price.*

This result follows from Proposition A.1 in the case without a deadline and Corollary 3 in the case with a deadline. These results demonstrate that even when the seller holds multiple units, the optimal policy remains threshold-based: there exists a reservation price such that the seller accepts any offer above it.

In the no-deadline case, the concavity of the value function implies that the marginal value of inventory increases with each additional unit sold, and the optimal sell-all reservation price is characterized by the solution to equation (A.5). This threshold balances the trade-off between selling immediately and waiting for future offers. With a deadline, the same logic applies, but the reservation price becomes time-dependent, declining as the deadline approaches due to the diminishing option value of waiting.

Hypothesis 3 (Supply and Inventory) *For a given unit, the seller's reservation price weakly decreases as the remaining inventory increases.*

This result follows from the concavity of the value function with respect to inventory, which implies a diminishing marginal value for each additional unit. As inventory increases, the opportunity cost of selling one more unit decreases, making the seller more willing to accept lower offers.

Hypothesis 4 (Reservation prices and deadlines) *Reservation prices are lower when there is a deadline.*

Hypothesis 4 asserts that theoretically optimal reservation prices are lower when a deadline is present. We show that this is numerically true for the parameters in our experiment in Appendix Table A.1. To prove the result more generally would require showing that the marginal value of inventory under a deadline, $\Delta \mathcal{W}_{T-j}(n-i+1)$, is strictly less than its counterpart in the stationary (no-deadline) case, $\Delta \mathcal{W}(n-i+1)$, for all $T-j$ and $n-i$.

The economic intuition is straightforward: a deadline shortens the horizon over which sellers can wait for better offers, thereby reducing the option value of holding inventory. This compresses continuation values and flattens the value function, leading us to expect a lower marginal value of an additional unit. Formalizing this relationship across the full dynamic program is complex, and we leave a complete theoretical treatment for future work.

3 Experimental design

We conduct an experiment to explore individuals' selling behavior when they hold multiple indivisible units of a good and encounter buyers stochastically, each offering to purchase a limited number of units at a specified per-unit price. The experiment is designed to examine the sellers' dynamic decision-making when they face uncertainty about future trading opportunities. Participants were recruited through the online platform Prolific (www.prolific.com), and the experiment was conducted entirely online. To participate in the experiment, subjects had to reside in the United States and speak English fluently.

Each subject in our experiment is assigned to one of two dynamic selling environments: one with no deadline for selling inventory, or one in which a hard deadline of three stages is imposed. In both treatments, subjects complete eight rounds. Each round consists of a

sequence of ‘trading stages,’ where each stage corresponds to an interaction with one potential buyer. In each stage, subjects have the opportunity to exchange a subset of their inventory for lottery tickets, which determine their probability of receiving a final monetary payment. Units that are not traded yield no payoff. The only difference between the two treatments is that, in the deadline condition, any units not traded within three stages are forfeited and cannot be converted into lottery tickets.

Subjects begin the experiment by reading instructions that describe the structure of the decision problems they will face. They are then shown a series of examples illustrating the available information, the choices they must make, and the resulting payoffs in a single round. After reviewing the instructions and examples, subjects proceed to the main phase of the experiment. Screenshots of all instructions and examples are provided in Online Appendix D.

In each round, subjects are endowed with either 1, 2, or 4 units of the good, which we refer to as the subject’s *inventory*.⁴ Subjects are also informed about the characteristics of the stochastic *demand structure* they will face. Specifically, they are given information about the probability distribution of each buyer’s maximum demand, as well as the distribution of per-unit prices that buyers may offer. In all stages, price offers are drawn from a discrete uniform distribution over the integers 1 through 25. The subject’s initial inventory and the distribution of buyers’ maximum demand together define a *configuration* of the problem. We use eight distinct configurations (labeled A through H and summarized in Table 1), which are presented to each subject in random order. This design spans a rich set of configurations, which is important in dynamic environments where continuation values and trading opportunities differ across stages, generating heterogeneous decision problems.

In each stage, a subject’s task is to specify reservation prices at which she is willing to sell her inventory. A reservation price must be set for each feasible quantity, up to the minimum of the subject’s inventory and the maximum possible demand. These reservation prices must be between 1 and 25 (the minimum and maximum possible price offers) and are required to be weakly increasing in quantity. Without this monotonicity restriction, subjects’ reported reservation prices would not uniquely determine the quantity traded, requiring an additional

⁴To make the decision problem more natural for participants, it is framed as selling lobsters; a good that (i) has high value, (ii) naturally spoils (providing a rationale for the stochastic termination described below), and (iii) is likely to be familiar to most subjects.

rule to determine how many units are sold. For example, one could specify that the largest such quantity is traded, but subjects would then need to understand this additional feature of the mechanism and the incentives it implies. Because non-monotone supply schedules are not optimal in the theoretical environment we study, we instead require choices to respect monotonicity, which allows us to implement the theoretically relevant decision problem without introducing this additional complexity.

After the subject submits her reservation prices, a quantity demanded and a price offer are randomly drawn according to the parameters of the current configuration. The computer then identifies the highest reservation price that is weakly less than the realized price offer. The subject sells either the quantity demanded or the quantity associated with this reservation price, whichever is smaller. Her earnings from the stage equal the product of the price offer and the number of units sold.

Prices are denominated in lottery tickets, where each ticket represents a 1% chance of receiving a bonus payment of \$10. Many of the theoretical predictions discussed in Section 2 rely on the assumption of risk neutrality. Under expected utility theory, even risk-averse subjects behave as if they are risk neutral in this context, since their choices influence only the probability of receiving a fixed prize (Roth and Malouf, 1979).⁵

Configuration	Inventory	Demand Structure	Distribution				Relevant Hypotheses
			1 unit	2 units	3 units	4 units	
A	1	1	100%	0	0	0	2, 3, 4
B	2	2	0	100%	0	0	1, 2, 3, 4
C	2	3	90%	10%	0	0	2, 3, 4
D	2	1	100%	0	0	0	3, 4
E	4	4	0	0	0	100%	1, 2, 4
F	4	5	90%	0	0	10%	2, 4
G	4	2	0	100%	0	0	3, 4
H	4	3	90%	10%	0	0	3, 4

Table 1: Configurations of seller’s inventory level and the distribution of buyer’s maximum demand.

⁵While this approach is standard, our empirical analysis focuses on comparative responses across treatments, which are less sensitive to the exact curvature of preferences.

Configurations and hypotheses The configurations are designed to serve as the basis for testing our hypotheses. For instance, Configurations B and E allow us to determine whether the reservation price remains constant when the seller holds multiple units and there are no demand restrictions (Hypothesis 1). A, B, C, E, and F allow us to test whether the reservation price to sell all units equals the single-unit reservation price (Hypothesis 2). To test whether higher inventory levels lead to lower reservation prices (Hypothesis 3), we fix the demand structure and vary the inventory level using configurations (A, D), (B, G), and (C, H) for demand structures 1, 2, and 3, respectively. Finally, all configurations serve as the basis for testing whether deadlines decrease reservation prices (Hypothesis 4).

Rounds and trading stages Each round consists of different trading stages. At the beginning of each trading stage, the subject needs to determine the minimum price per unit they are willing to accept to sell any given number of units (i.e., a reservation price), up to the minimum of the number of units they hold and the maximum demand that can appear. For instance, a subject in Configuration F starts with an inventory of 4 units, and the buyer may demand up to four units. Thus, such a subject would need to determine the minimum price per unit to trade 1 unit, 2 units, and so on, up to 4 units. A subject in Configuration H starts with an inventory of 4 units, but faces a maximum demand of 2 units. Thus, she only needs to choose the minimum price per unit to sell 1 and 2 units. Figure 2 shows the interface subjects see when facing Configuration H.

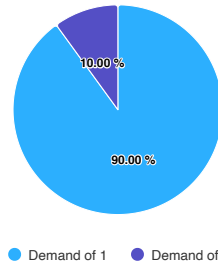
Round 1, Stage 1

You currently have a stock of **4 lobsters** and so far, you have 0 lottery tickets in this round.

There is a 20% chance that your lobsters will spoil after this stage.

In this round, there is a 90% chance that the number of lobsters demanded is 1 and 10% chance that it is 2. That means that given your stock, the maximum you could sell is 2.

Demand Chances



In the boxes below, please list the minimum number of lottery tickets per lobster you would be willing to accept to sell each number of lobsters up to 2. Remember that the numbers you list must be between 1 and 25, and the minimum number of tickets must increase with the number of lobsters.

Minimum price per unit to sell 1 unit

Minimum price per unit to sell 2 units

Next

Figure 2: Reservation prices - No deadline

Feedback After choosing these reservation prices, subjects observe the realization of the random price per-unit and the maximum number of units demanded by the buyer. The subject is informed of how their reservation prices have been used: given the realized price offer, they sell the minimum of the amount demanded and what they are willing to sell.

After each trade, the subject is informed about the terms of the trade (i.e., how many units were sold and at what price), the number of units remaining, if any, and the total probability accumulated. Figure 3 shows the feedback outcome for a subject who did not sell any units, whereas Figure 4 illustrates the case of a subject who sold all units in a given round.

Results: Round 1, Stage 1

In this stage, the buyer offered 21 lottery tickets per lobster. On the previous page, you reported that the minimum you would accept to sell one lobster is 25. Because the random price is below this, you have not agreed to sell any lobsters at this price.

This stage's random demand was 1. That means that you sold 0 lobsters and received 0 lottery tickets this stage.

You have accumulated a total of 0 lottery tickets through sales in this round.

Next

Figure 3: Stage results - No sales

Results: Round 2, Stage 1

In this stage, the buyer offered 22 lottery tickets per lobster. That is higher than the highest minimum acceptable price you reported on the previous page, so you have agreed to sell up to 4 lobsters at this price.

This stage's random demand was 4. That means that you sold 4 lobsters and received 88 lottery tickets this stage.

You have accumulated a total of 88 lottery tickets through sales in this round.

Next

Figure 4: Stage results - Sales

Termination If the subject does not sell all the units she holds in the current trading stage, there is an 80% chance that the subject continues onto the next stage; otherwise, the decision process ends. The subject is informed about the result of the continuation process after each stage. When the decision process ends, either because the subject depleted her whole inventory or because the decision process was terminated stochastically, the subject is informed about the total probability accumulated. Figure 5 illustrates a case in which the computer terminates the process as a result of an unfavorable draw.

Continuation: Round 4, Stage 1

The computer randomly rolled a 96, which is higher than 80. That means that this round is over.

Next

Figure 5: Continuation - Random termination

In the decision process with deadlines, the process can also terminate if the subject reaches the deadline without selling all of her inventory. Figure 6 shows the feedback presented to subjects when the deadline is reached and all unsold units are dissipated.

Continuation: Round 3, Stage 3

All of your lobsters were guaranteed to spoil after the last stage, so this round is over. You accumulated a total of 0 lottery tickets in this round, so if this round is chosen to be the one that counts, you have a 0% of receiving a bonus payment of \$10.



Figure 6: Continuation - Deadline reached

Payoffs At the end of the experiment, one configuration is randomly chosen to determine the payoffs. Each configuration has an equal chance of being selected. Subjects are notified of the chosen decision type after all decisions have been completed.

Data The experiment was carried out over a three-day period in September 2024. 201 subjects completed the experiment—101 in the treatment without deadlines and 100 with deadlines. We report summary statistics of demographic variables in Appendix Table C.1. All subjects received the baseline payment for completion of \$3.60, and 65 subjects received the bonus payment of \$10. The median time for completion was under 18 minutes. We report the distribution of median decision times for reporting reservation prices in the first stage of each search problem in Appendix Figure C.1. Table C.2 shows a fixed-effects regression of how decision times vary with problem characteristics.

4 Results

As discussed in our preanalysis plan, our empirical analysis focuses on data from the first stage of trade.⁶ This is because the set of subjects contributing data from later stages is en-

⁶The regression reported in Table 3 matches our pre-analysis plan exactly. The regression reported in Table 2 differs slightly from what was pre-registered, as discussed in Footnote 7. Relative to the pre-registration, the regression reported in Table 4 adds Response Time as a control—the version of the table without response times can be found in Appendix Table C.4. No analysis from the pre-analysis plan was omitted in the paper. All other statistical analysis should be viewed as exploratory.

dogenous: those who report higher reservation prices in the first stage are more likely to provide data in subsequent stages, raising concerns about selection bias. Each subject contributes 18 reservation price observations, corresponding to the total number of reservation prices that must be set for the maximum number of units allowed by each configuration.

In the four sections below, we compare empirical reservation prices to their theoretical predictions. First, we graphically compare the empirical and theoretical supply curves to identify patterns that allow us to test the first four hypotheses. Second, we examine the effect of increasing inventory while holding demand constant. Third, we analyze the impact of introducing a deadline while keeping both inventory and demand fixed. Finally, we evaluate the characteristics of decision problems that lead to deviations from theoretical predictions.

Because the magnitude of the predicted effect of inventory varies across demand structures, and the magnitude of the predicted effect of deadlines varies across both demand structures and inventory levels, we pool the analysis by focusing on a single summary variable, which we refer to as PE (for “predicted effect”). These predicted effects reflect the theoretical impact of our treatment variables—inventory levels and the presence of a deadline—on reservation prices. For example, suppose that for a fixed unit demand, the reservation price to sell the unit is 13 when the seller holds one unit of inventory, and 9 when holding two units. In this case, we define the variable PE Inv as -4 . If subjects’ responses to increased inventory and deadlines align with theoretical predictions on average, the estimated coefficients on PE Inv and PE DL should be equal to one. Coefficients less than one indicate an *under-reaction* to the treatment, while coefficients greater than one indicate an *over-reaction*.

4.1 The effect of demand restrictions and supply curves

In this section, we compare the theoretical supply curves implied by the reservation values with their empirical counterparts. The analysis distinguishes between environments with and without deadlines to examine how time constraints influence sellers’ willingness to sell. For clarity, we present the latter case in the Appendix, as the patterns remain qualitatively unchanged relative to our hypotheses.

Figure 7 presents results for configurations A, B, C, and D. In all cases except configuration A — where sellers hold a single unit — sellers are endowed with two units, while the demand

distribution varies across configurations. Demand structures 1, 2, and 3 are considered. Across most configurations — except for D — the empirical supply curves lie below their theoretical counterparts, indicating that sellers tend to accept lower prices than predicted. Notably, the gap between empirical and theoretical curves narrows as the variance of the demand distribution increases, as observed in configuration C. Furthermore, the empirical reservation price for the second unit more closely aligns with the theoretical prediction than that of the first unit, suggesting greater adherence to theory when selling subsequent units.

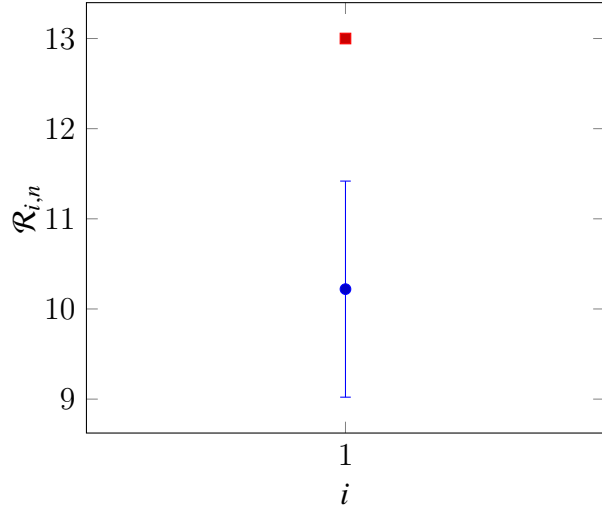
In turn, Figure 8 displays the theoretical and empirical supply curves for configurations E, F, G, and H under the no-deadline condition. In all configurations, the inventory level is fixed at four units, while the demand distribution varies across settings. Here, we consider demand structures 2, 3, 4, and 5. The empirical supply curve lies above the theoretical one for configurations with non-zero variance (F and H). In contrast, for configurations E and G, the empirical supply curve crosses the theoretical curve from below. Unlike the case where the inventory is fixed at two units, the empirical reservation price aligns more closely with the theoretical prediction for the first unit than for subsequent units.

Configuration A corresponds to the classic search problem with a single unit. Thus, behavior in this search problem serves as a baseline point of comparison between our experiment and the existing literature.

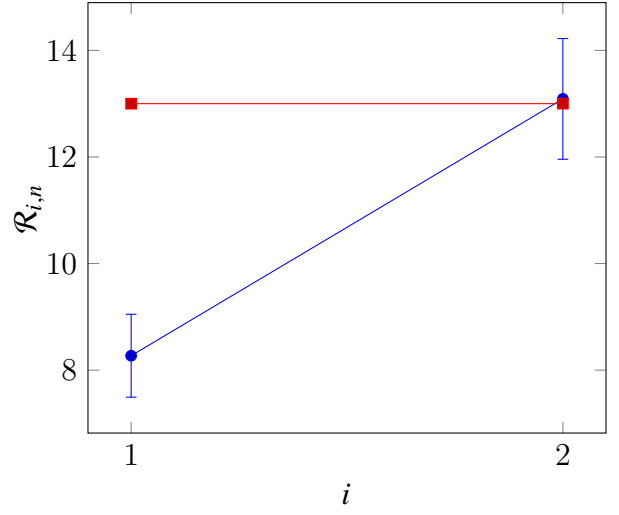
Result 1 *When the seller holds a single unit and buyers have unitary demand, empirical reservation values are unequivocally lower than the theoretically predicted value.*

Our findings therefore reinforce those of Sonnemans (1998) within our experimental framework; particularly given that, under expected utility maximization, subjects should behave as if they are risk neutral.

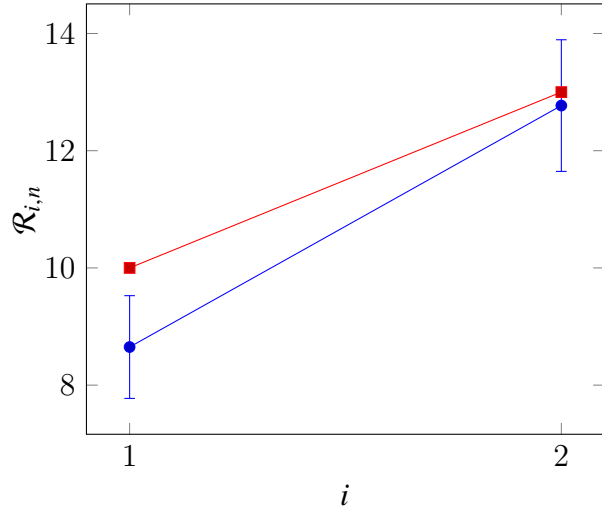
To test Hypothesis 1, we use configurations B and E. In configuration B, the seller holds two units and encounters buyers who demand exactly two units, while in configuration E, the seller holds four units and encounters buyers who demand exactly four units. In both configurations, we reject the hypothesis of constant reservation values. While the evidence clearly rejects the hypothesis of constant reservation values, it is reasonable to object that our design *requires* reservation prices to be weakly increasing. Thus, any noise in decision-making



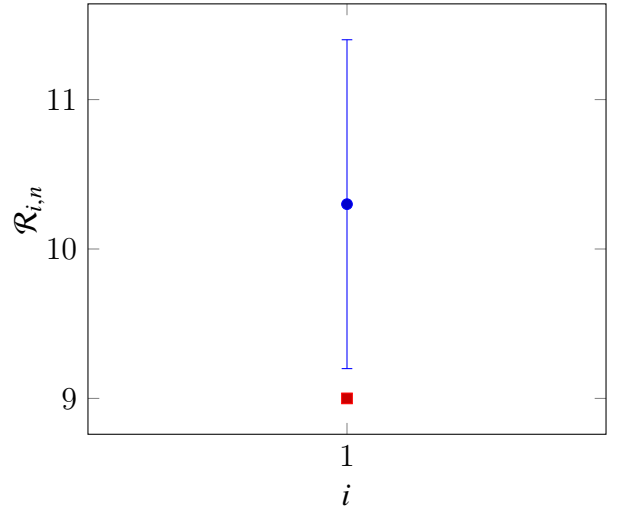
(a) Configuration A. Inventory: one unit. Demand distribution: $\alpha_1 = 1$.



(b) Configuration B. Inventory: two units. Demand distribution: $\alpha_2 = 1$.

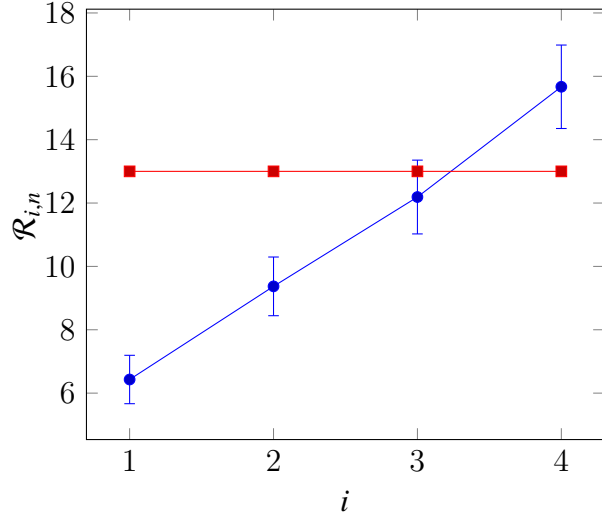


(c) Configuration C. Inventory: two units. Demand distribution: $\alpha_1 = 0.9$ and $\alpha_2 = 0.1$.

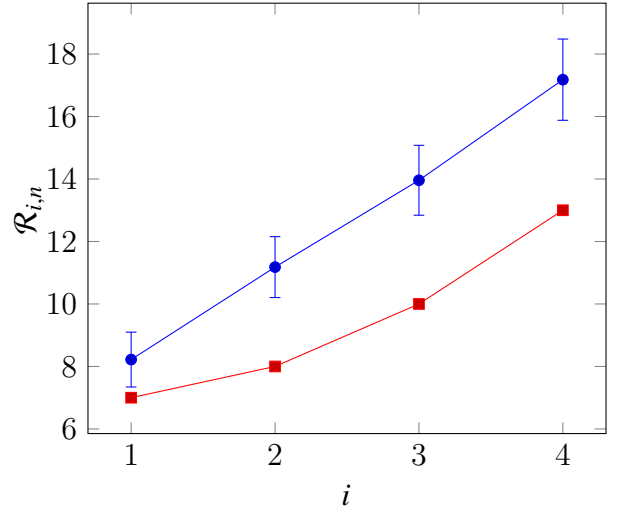


(d) Configuration D. Inventory: two units. Demand distribution: $\alpha_1 = 1$.

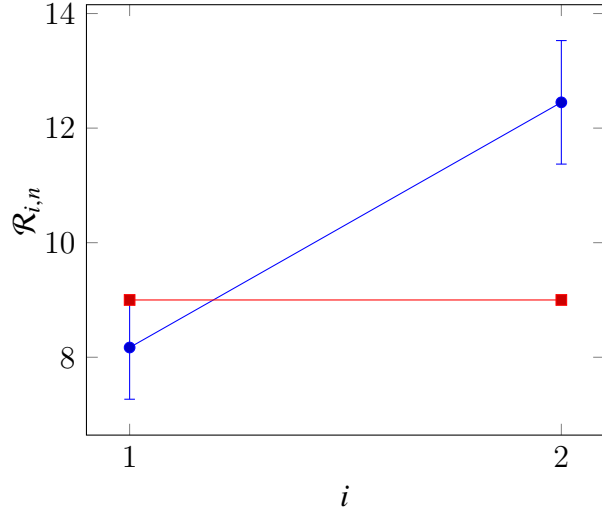
Figure 7: Comparison of Configurations A, B, C, and D with no deadline. Theoretically predicted supply curves are shown in red, while empirical supply curves with confidence intervals are shown in blue. The empirical supply curves reflect average reservation prices for each unit in the first stage of the search process. Error bars indicate 95% confidence intervals, computed using clustered standard errors from a regression of reservation prices on unit dummies.



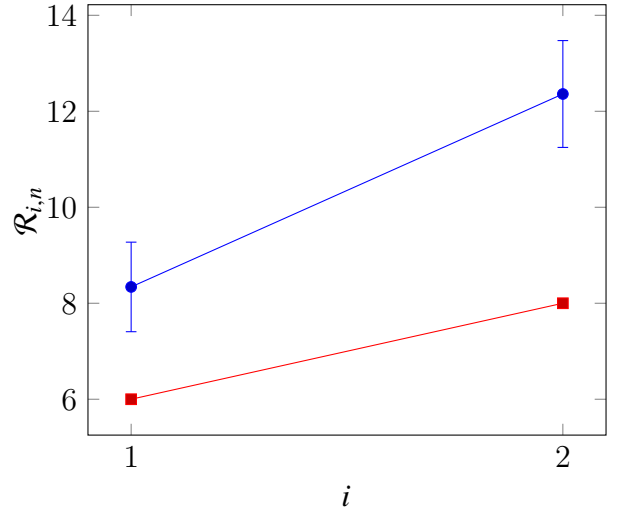
(a) Configuration E. Inventory: four units.
Demand distribution: $\alpha_4 = 1$.



(b) Configuration F. Inventory: four units.
Demand distribution: $\alpha_1 = 0.9$ and $\alpha_4 = 0.1$.



(c) Configuration G. Inventory: four units.
Demand distribution: $\alpha_2 = 1$.



(d) Configuration H. Inventory: four units.
Demand distribution: $\alpha_1 = 0.9$ and $\alpha_2 = 0.1$.

Figure 8: Comparison of Configurations E, F, G, and H with no deadline. Theoretically predicted supply curves are shown in red, while empirical supply curves with confidence intervals are shown in blue. The empirical supply curves reflect average reservation prices for each unit in the first stage of the search process. Error bars indicate 95% confidence intervals, computed using clustered standard errors from a regression of reservation prices on unit dummies.

leads to a rejection of the hypothesis. As an additional point of comparison, in Appendix Table C.5 we estimate the slope of the supply functions for Configurations B, C, E, and F using Subject $\times$ Configuration fixed effects. We can then test whether supply functions are flatter when demands are unrestricted (Configurations B and E). We find that in both cases, the estimated slopes are *higher* for the configurations with unrestricted demand, confirming Result 2.

Result 2 *When sellers hold more than one unit and face buyers with unrestricted demands, reservation values are increasing, not constant. Thus, we reject Hypothesis 1.*

Configurations A, B, C, E, and F are used to test Hypothesis 2. For this purpose, we focus on the reservation price for selling all units in each configuration. These values are 10.22, 13.09, 12.77, 15.67, and 17.18, respectively. To assess the statistical significance of these differences, we regress reservation prices on configuration dummies and test whether the coefficients on the dummies are jointly equal to zero. This null hypothesis is rejected ($p < 0.01$; full results available upon request).

Result 3 *The reservation price to sell all units depends on the inventory level and is generally different from the reservation price in the single-unit case. Thus, we reject Hypothesis 2.*

Figures B.1 and B.2 in the Appendix B replicate Figures 7 and 8, respectively, for the case with a deadline. Most of the qualitative results remain the same in this case. The empirical and theoretical supply curves for Configuration C are nearly identical, though we believe this is likely due to random variation rather than any special feature of the configuration. The empirical supply curves for Configurations F-H are uniformly above their theoretical counterparts, indicating, as discussed further below, that subjects react less strongly to deadlines than theory would predict.

4.2 The effect of increasing inventory

We now consider how changes in inventory levels influence reservation prices. To test Hypothesis 3, we estimate the fixed-effects model presented in Equation 1:

$$\text{Res. Price}_{icu} = \beta_0 + \beta_1 \text{High Inv}_c \times \text{PE Inv}_{cu} + \sum_{k=1}^5 \delta_k \mathbb{1}\{\text{Demand-Unit}_{cu} = k\} + \alpha_i + \varepsilon_{icu}. \quad (1)$$

The indices are defined as follows: i denotes the subject, c the configuration, and u the unit number. The dependent variable, Res. Price_{icu} , represents the reservation price reported by subject i for unit u under configuration c . The variable High Inv_c equals 1 for configurations D, G, and H. The demand-unit indicators $\mathbb{1}\{\text{Demand-Unit}_{cu} = k\}$ control flexibly for each unit within the demand distribution that is used in configuration c .⁷ The variable PE Inv_{cu} is explained at the beginning of this section. The parameter of interest is β_1 , and we estimate Equation 1 separately for the deadline and no-deadline treatments.⁸

⁷In our pre-analysis plan, we wrote that we would estimate this regression with demand indicators instead of demand-unit indicators. We believe that this specification is better because comparisons are made within-rather than across-unit. Nevertheless, we report the results of the pre-registered regression in Appendix Table C.3. The qualitative results are largely the same. The only notable difference is that the coefficient on Higher Inventory $\times$ Predicted Effect is statistically significant, albeit still small and with a sign opposite to that of the theoretical prediction.

⁸For this regression, we will use only data from Configurations A, B, C, D, G, and H because these are the configurations with variation in inventory for fixed demand structures. Thus, each subject contributes 10 observations to this regression.

	(1)	(2)	(3)	(4)
	Res. Price	Res. Price	Res. Price	Res. Price
Higher Inventory	-0.28 (0.28)		0.014 (0.23)	
Higher Inventory $\times$ Predicted Effect		0.067 (0.068)		-0.014 (0.035)
Constant	10.4*** (0.32)	10.4*** (0.32)	9.78*** (0.33)	9.76*** (0.32)
Type	No Deadline	No Deadline	Deadline	Deadline
Observations	1010	1010	1000	1000

Notes: Linear regression with subject and demand-unit fixed effects and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Effect of inventory levels on reservation prices

Table 2 summarizes the effects of increased inventory levels while holding the demand structure constant. Columns (1) and (3) report the results of regressing reservation prices on a dummy indicating higher inventories *without* accounting for how the theoretical predictions vary based on configuration. Columns (2) and (4) interact the dummy for higher inventory with the theoretically predicted effect of the higher inventory. The interaction coefficient is the ratio of the empirical to the predicted effect.

Across all specifications, we find no statistically significant effect of increased inventory. In the no-deadline treatment, the estimated effect has the predicted sign, but the 95% confidence interval rules out effects greater than 20% of the theoretically predicted magnitude. In the deadline treatment, the sign of the effect is contrary to theoretical predictions: increased inventory slightly *raises* reservation prices, although the effect is statistically insignificant.

Result 4 *Inventory levels do not have a statistically significant effect on how sellers set reservation prices. Thus, we reject Hypothesis 3.*

4.3 The effect of deadlines

We next examine how the introduction of deadlines affects reservation prices, providing a direct test of Hypothesis 4. If subjects anticipate the possibility of being unable to continue the selling process, they may become more eager to sell, leading them to set lower reservation prices. To test this, we estimate the following random-effects model:

$$\text{Res. Price}_{icu} = \beta_0 + \beta_1 \text{DL}_i \times \text{PE DL}_{cu} + \sum_{k=1}^{18} \gamma_k \mathbb{1}\{\text{Configuration-Unit}_{cu} = k\} + \alpha_i + \varepsilon_{icu}. \quad (2)$$

Variable definitions follow the same notation as above. DL_i equals one if subject i is assigned to the deadline treatment. The term PE DL_{cu} is defined earlier in the section. The configuration-unit indicators $\mathbb{1}\{\text{Configuration-Unit}_{cu} = k\}$ control flexibly for each unit within each configuration. Standard errors are clustered at the subject level. This regression includes all configurations; each subject contributes 18 observations, one for each reservation price decision across all configuration-unit combinations. The coefficient of interest is β_1 .

	(1)	(2)
	Res. Price	Res. Price
Deadline	-0.37 (0.53)	
Deadline $\times$ Predicted Effect		0.022 (0.077)
Constant	10.2*** (0.51)	10.0*** (0.41)
Configuration-unit FE	Yes	Yes
Observations	3618	3618

Notes: Linear regression with subject random effects, configuration-unit fixed effects, and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: The effect of deadlines on reservation prices

Table 3 reports the results of estimating Equation 2. Because this is a between-subject comparison, each subject contributes 18 reservation price observations; one for each configuration-unit pair. In contrast to the within-subject analysis of inventory effects, the deadline effect reflects between-subject differences across treatment groups.

Column (1) shows a regression using a dummy for deadline assignment. Theory predicts a negative coefficient, as deadlines should induce lower reservation prices due to the risk of forfeiting unsold units. However, the estimated effect is near zero and statistically insignificant. Column (2) includes the interaction with the theoretically predicted effect. The estimate suggests that the empirical effect of deadlines is only 2% of the predicted magnitude.

We summarize our findings in the following result.

Result 5 *Deadlines do not have a statistically significant effect on how sellers set reservation values. Thus, we reject Hypothesis 4.*

4.4 The determinants of deviations from theory

Finally, we investigate which characteristics of the search problem drive deviations from theoretical predictions. For this analysis, we estimate the following fixed-effects model:

$$\text{Deviation}_{icu} = \beta_0 + \mathbf{X}'_{icu}\beta + \alpha_i + \varepsilon_{icu}, \quad (3)$$

The dependent variable, Deviation_{icu} , measures the difference between the observed and the theoretically predicted reservation price. We estimate the model using both the raw difference and the absolute deviation as dependent variables. The matrix $\mathbf{X}_{icu}$ includes current inventory, unit number, average demand level, search number (a proxy for experience), and response time in seconds. The unit number refers to the position in the sequence of sales within a configuration; for instance, under configuration H, the seller has an inventory of 4 and faces buyers who may demand up to two units each. This seller must set separate reservation prices for the first and second unit sold—these correspond to unit numbers 1 and 2. The average demand level is the expected number of units a buyer demands, calculated as the probability-weighted average across the demand distribution (e.g., under demand structure 5, average

demand is $0.9 \times 1 + 0.1 \times 4 = 1.3$). The coefficients β capture the impact of these characteristics on deviations.

This regression includes data from all configurations, resulting in each subject contributing 18 observations. By pooling these observations across subjects, we aim to identify systematic patterns that explain deviations from theoretical benchmarks.

	(1)	(2)	(3)	(4)
	Deviation	Deviation	Abs(Deviation)	Abs(Deviation)
Inventory	2.02*** (0.094)	2.02*** (0.095)	0.38*** (0.11)	0.39*** (0.11)
Unit Number	1.57*** (0.13)	1.57*** (0.13)	0.44*** (0.087)	0.45*** (0.087)
Average Demand	-2.66*** (0.10)	-2.66*** (0.10)	-0.16 (0.099)	-0.16 (0.099)
Search Number	0.031 (0.051)	0.029 (0.053)	0.098*** (0.036)	0.092** (0.039)
Response Time		-0.00023 (0.0020)		-0.00088 (0.0013)
Constant	-3.18*** (0.42)	-3.17*** (0.43)	3.14*** (0.32)	3.18*** (0.34)
Observations	3618	3618	3618	3618

Notes: Linear regression with subject fixed effects and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Deviations from the theoretical benchmark

Table 4 reports the relationship between structural features of the problem and deviations from theoretically optimal reservation prices. Columns (1) and (2) use the raw deviation as the dependent variable, while Columns (3) and (4) use the absolute deviation.

Columns (1) and (2) show that raw deviations are positively associated with inventory and unit number and negatively associated with average demand. It is straightforward to see

these effects from Figures 7 and 8. Subjects make minimal changes to their reservation prices in response to higher inventory or demand. Because higher levels of inventory (demand) are theoretically predicted to decrease (increase) reservation prices, the empirical deviation variable increases (decreases). In the case of unit number, Figures 7 and 8 show that subjects tend to produce steeper supply functions than theory would predict, leading to a positive correlation between deviation and unit number.

Columns (3) and (4) focus on absolute deviations and provide further insight into the magnitude of departures from theory. Inventory and unit number remain positively associated with deviation size. Notably, the positive coefficient on search number (0.098 in Column 3) suggests that deviations increase modestly with the search number, contrary to expectations of better decision-making with experience. This may reflect cognitive fatigue or declining engagement as subjects progress through repeated decision tasks.

Perhaps surprisingly, response times are not associated with deviations or their absolute value. Thus, while subjects spend more time on some tasks than others, it does not appear that this extra time is associated with making choices that are closer to the rational benchmark.

To determine whether responsiveness to our treatment variables may be heterogeneous, we repeat our main empirical analysis from Sections 4.2-4.4 for subjects with high and low Cognitive Reflection Test (CRT) scores in Appendix Tables C.6-C.8. We do not find systematic differences across CRT groups.⁹

Finally, we evaluate the payoff consequences of the deviations we document. For each search problem, we calculate the expected loss from the subject’s reported first-stage price schedule relative to the rational benchmark, assuming that reservation prices in subsequent stages are optimal.¹⁰ The mean loss according to this metric is 1.34 lottery tickets, corresponding to 13.4 cents in expected earnings. After first averaging losses within subject, the median loss is 1.12 tickets. 42% of subjects have average losses of less than 1 ticket, while 16% have average losses of more than two tickets. Thus, despite the substantial size of observed deviations, their payoff consequences are relatively modest.

⁹We also completed this analysis separately for subjects with particularly slow or fast response times without finding systematic differences. Results are available upon request.

¹⁰Focusing only on reservation prices from the first stage is consistent with the rest of our analysis and avoids the complication that earlier choices affect which later decision nodes are reached.

5 Alternative Behavioral Benchmarks

Our experimental results show that behavior is not well-explained by the rational benchmark provided by Carrasco and Smith (2017). In this section, we explore two alternative models for behavior in dynamic search problems. The first assumes that decision makers may not consider the continuation value inherent in holding an inventory far into the future. The second proposes that in the face of a complex environment, subjects follow simple heuristic decision rules that, while suboptimal, are nevertheless stable across environments.

5.1 Finite Look-ahead

One potential explanation for the weak response of reservation prices to inventory levels and deadlines is that subjects do not fully account for the continuation value of future trading opportunities. Experimental evidence from dynamic decision problems documents substantial heterogeneity in subjects' ability to plan ahead (Carbone and Hey, 2004). More directly, Baumann et al. (2023) find that when the search horizon changes in sequential search problems, subjects do not adjust their stopping thresholds as much as would be optimal. Motivated by this literature, we consider a finite-look-ahead benchmark in which sellers optimize dynamically but only over a limited planning horizon.

Let $h \in \{0, 1, 2, \dots\}$ denote the number of future trading opportunities incorporated into the seller's decision problem. Let $W^{(h)}(n)$ denote the value of holding inventory n when the seller looks ahead at most h future opportunities. Define recursively

$$W^{(0)}(n) = 0$$

and, for $h \geq 1$,

$$W^{(h)}(n) = \beta E \left[\max_{y \leq \min\{n, Q\}} \left\{ Py + W^{(h-1)}(n - y) \right\} \right].$$

The marginal value of the u -th unit in an inventory of size n is

$$R_{u,n}^{(h)} = W^{(h)}(n - u + 1) - W^{(h)}(n - u), \quad u = 1, \dots, n.$$

As h increases, the seller incorporates additional future trading opportunities into the decision problem. The benchmark with $h = 0$ corresponds to complete myopia, while the limit as $h \rightarrow \infty$ coincides with the optimal policy implied by the model.

In Table 5, we evaluate how well the finite look-ahead model fits our data. The table reports the root mean squared error of each horizon level h if that horizon level is applied to all subjects. It also reports the proportion of subjects whose behavior is best predicted by a given horizon level according to mean squared error. We note that for subjects in the deadline treatment, all horizon levels of 2 or higher coincide with the rational benchmark (which we label as $h = \infty$) because in the initial stage, there are only two following periods.

		$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = \infty$
No deadline	RMSE	11.68	8.70	7.30	6.60	6.32	6.14
	Best fitting (%)	8.9	0.0	5.0	0.0	9.9	76.2
Deadline	RMSE	11.39	8.49				7.18
	Best fitting (%)	11.0	2.0				87.0

Table 5: Fit of Finite Look-Ahead Benchmarks. RMSE is the root mean squared error between observed and predicted reservation prices. Best fitting reports the percentage of subjects for whom each look-ahead horizon minimizes mean squared error. Ties are assigned to the longer look-ahead horizon. In the deadline treatment, all horizons $h \geq 2$ generate the same predictions as the fully rational benchmark; this common benchmark is reported in the $h = \infty$ column.

The results show that the finite look-ahead model does not provide a substantially better fit for behavior in our experiment than the rational benchmark. For the pooled data from both the deadline and no deadline treatments, the rational benchmark fits the data better than any shorter horizon. Turning to individual classifications, over 75% of subjects in *both* treatments have behavior that is best fit by the rational benchmark. Thus, there is not strong evidence that a model of finite look-ahead explains behavior in our experiment.

5.2 Heuristic Choice & Stable Supply Schedules

A second possibility is that subjects do not compute state-contingent reservation prices at all. Instead, they may rely on simple pricing rules that remain largely invariant across environments. This possibility is broadly consistent with evidence that uncertainty about optimal strategies can attenuate responses to changes in objective parameters and that decision

makers adopt simple and suboptimal decision rules in complex environments (Banovetz and Oprea, 2023; Enke and Graeber, 2023). To investigate this possibility, we estimate two simple subject-specific stable supply schedules.

The first candidate, which we call the linear stable schedule, assumes that supply schedules are affine with subject-specific intercept and slope terms and that they do not otherwise depend on the environment’s characteristics. Thus, each supply schedule satisfies

$$R_{icu} = a_i + b_i(u - 1) + \varepsilon_{icu},$$

where R_{icu} denotes the reservation price chosen by subject i for unit u in configuration c . Here, a_i is a subject-specific baseline reservation price, and b_i represents how reservation prices increase for later units. The unit index u is ordered from the first unit offered for sale to the last unit available in the seller’s inventory.

Our second candidate, the nonparametric stable schedule, also depends only on the unit number but does not impose linearity. Thus,

$$R_{icu} = \alpha_{iu} + \varepsilon_{icu}$$

where R_{icu} is as above and α_{iu} is subject i ’s mean reservation price for unit u .

Importantly, neither stable supply schedule depends on inventory levels or demand distributions. Instead, subjects are assumed to follow a pricing rule whose only variation across environments is due to idiosyncratic mean-zero noise. The underlying idea behind this benchmark is that subjects may rely on a simple heuristic pricing schedule instead of calculating continuation values for each decision problem.

We compare the out-of-sample predictive performance of these stable supply schedules with that of the rational benchmark. Because the stable-schedule benchmarks contain subject-specific parameters that must be estimated from observed choices, we evaluate them using cross-validation. For the linear stable schedule, we take each configuration in turn, estimate a linear relationship between reservation price and unit number for each subject using the other seven configurations, and use the estimated relationship to predict all reservation prices in the held-out configuration. For the nonparametric stable schedule, we predict each subject’s

reservation price for a given unit using the mean of that subject’s reservation prices for the same unit in the other configurations. Table 6 reports the root mean squared prediction error for each stable-schedule benchmark alongside that of the rational benchmark.

	No deadline	Deadline	Pooled
Optimal dynamic model	6.14	7.18	6.68
Linear stable schedule	4.25	4.31	4.28
Nonparametric stable schedule	4.38	4.44	4.41

Table 6: Root Mean Squared Error for Stable Supply Schedules. The predictions for the stable schedules are constructed by estimating the supply schedule having left out the configuration being predicted.

The results show that the simple linear stable supply schedule predicts choices better than the nonparametric stable supply schedule and substantially better than the rational benchmark. This suggests that subjects’ reservation prices have a stable, low-dimensional structure that is shared across configurations. This pattern is consistent with subjects using a simple heuristic to determine their pricing schedule rather than fully adapting their reservation prices to each new environment.

6 Concluding Remarks

This paper provides empirical evidence from laboratory experiments that test predictions about dynamic seller pricing in multi-unit search models. We test the theoretical work of Carrasco and Smith (2017) by examining how reservation prices evolve in a decentralized, multi-unit setting with stochastic buyer arrivals, focusing on inventory levels and deadlines. Contrary to theoretical predictions, we find that neither inventory nor deadlines significantly affect pricing behavior, suggesting that theoretical models should be enriched.

To investigate these deviations from the rational benchmark, we evaluate two potential behavioral explanations. A model of finite look-ahead does not fit the data better than the rational benchmark, suggesting that the results cannot be explained by subjects’ short planning horizons. In contrast, estimated subject-specific stable supply schedules predict better than the rational benchmark out of sample. This suggests that subjects may be relying on simple

and relatively stable pricing rules rather than solving for the optimum supply schedule as the characteristics of their decision problem change.

The deviations we document suggest that sellers that rely on individual judgment may price suboptimally, potentially creating a role for decision aids or automated pricing systems. At the same time, stable supply schedules should be thought of as a descriptive benchmark rather than a full explanation of the source of behavior. While our payment mechanism allows us to rule out standard risk aversion as an explanation for behavior, choices could still be driven by non-expected-utility preferences, including nonlinear probability weighting. For this reason, identifying tractable models of such preferences in dynamic multi-unit search environments remains an important direction for future research.

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Appendix

A Derivation of Theoretical Results

A.1 Values and reservation prices with no deadlines

When there are no deadlines, the environment is stationary, and we can drop the stage sub-index t . Throughout Appendix A, we express prices and accumulated earnings in probability units: a price of p lottery tickets in the experiment corresponds to $p/100$ in the notation below. Thus, π denotes the accumulated probability of receiving the high utility payoff $\bar{u}$ rather than $\underline{u}$, and selling y units at price P increases this probability by Py .

Let n be the *post-trade inventory* so that the *continuation value function* obeys

$$\mathcal{V}(n, \pi) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} \mathcal{V}(n - y, \pi + Py) \right) + (1 - \beta) \mathcal{E}(\pi). \quad (\text{A.2})$$

where $\mathcal{E}(\pi) = \underline{u} + \pi(\bar{u} - \underline{u})$. As we now formally show in our next result, the seller maximizes optionality and, consequently, the expected utility of the lottery. More precisely, the seller maximizes the probability of receiving the utility payoff of $\bar{u}$.

Proposition A.1 (Values without deadline) *When $T \rightarrow \infty$ then $\mathcal{V}(n, \pi) = \underline{u} + (\pi + \mathcal{W}(n))(\bar{u} - \underline{u})$, where $\mathcal{W}(n)$ solves*

$$\mathcal{W}(n) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}(n - y)) \right). \quad (\text{A.3})$$

Furthermore, the reservation price for selling $1 \leq i \leq n$ units equals

$$\mathcal{R}_{i,n} = \Delta \mathcal{W}(n - i + 1) = \mathcal{W}(n - i + 1) - \mathcal{W}(n - i). \quad (\text{A.4})$$

which weakly decreases as inventory increases.

Proof of Proposition A.1

We will prove the result by a guess-and-verify procedure for general utility payoffs $\bar{u} > \underline{u}$ of the lottery, as well as probability π . Substituting $\mathcal{V}(n, \pi) = \underline{u} + (\pi + \mathcal{W}(n))(\bar{u} - \underline{u})$ into (A.2) yields:

$$\underline{u} + (\pi + \mathcal{W}(n))(\bar{u} - \underline{u}) = \beta(\bar{u} - \underline{u})\mathbb{E}\left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}(n - y))\right) + \underline{u} + \pi(\bar{u} - \underline{u})$$

Canceling common terms yields $\mathcal{W}$ that satisfies (A.3). Since there is a unique fixed point for $\mathcal{W}$, it follows that $\mathcal{W}$ is a concave function, by Carrasco and Smith (2017). Consequently, the slopes $\Delta\mathcal{W}(n - i + 1) = \mathcal{W}(n - i + 1) - \mathcal{W}(n - i)$ represent the reservation prices for selling i units. As $\mathcal{W}$ is concave, reservation prices weakly decrease as inventory increases.

Corollary 1 (Single unit case or no demand restrictions) *When $n = 1$ or in the case without demand restrictions, the reservation prices are all equal to $\mathcal{R}$, which solves*

$$\mathcal{R} = \beta\mathbb{E}(\max\{P, \mathcal{R}\}) \tag{A.5}$$

while the continuation value function is $\mathcal{V}(n, \pi) = \underline{u} + (\bar{u} - \underline{u})(\pi + n\mathcal{R})$.

Proof of Corollary 1

Let $\mathcal{R}$ uniquely solve (A.5). When demand is unrestricted, substituting the candidate $\mathcal{W}(n) = n\mathcal{R}$ into the Bellman equation gives

$$\begin{aligned} \beta\mathbb{E}\left[\max_{y \in \{0, \dots, n\}} \{Py + (n - y)\mathcal{R}\}\right] &= \beta n\mathbb{E}[\max\{P, \mathcal{R}\}] \\ &= n\mathcal{R}. \end{aligned}$$

By uniqueness of the Bellman fixed point, $\mathcal{W}(n) = n\mathcal{R}$. Consequently, $\mathcal{W}(n - i + 1) - \mathcal{W}(n - i) = \mathcal{R}$, so all reservation prices equal $\mathcal{R}$. The continuation value is therefore

$$\mathcal{V}(n, \pi) = \underline{u} + (\bar{u} - \underline{u})(\pi + n\mathcal{R}).$$

The single-unit case follows by setting $n = 1$, since demand is always at least one.

A.2 Values and reservation prices with deadlines

We now assume that $T < \infty$ and so we need to keep track of the stage time to compute continuation values. However, we do not need to use it for state variables. As in our previous case without deadline, let n be the post-trade inventory. Then, since we have assumed that $\pi + \bar{p}n \leq 1$, we obtain that for any $j \in \{1, 2, \dots, T-1\}$:

$$\mathcal{V}_{T-j}(n, \pi) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} \mathcal{V}_{T-j+1}(n-y, \pi + Py) \right) + (1-\beta) \mathcal{E}(\pi). \quad (\text{A.6})$$

Naturally, at the final stage $t = T$ the continuation value is just the expected value of the lottery; thus, $\mathcal{V}_T(n, \pi) = \mathcal{E}(\pi) = \underline{u} + \pi(\bar{u} - \underline{u})$.

The following result is the exact analog to our Proposition A.1 for the case with a deadline T . The seller maximizes optionality and thus the probability of the high reward of $\bar{u}$. To wit, the value of holding inventory is the expected value of the lottery, as we specify in (A.7).

Proposition A.2 (Values with deadline) *For $j \in \{1, 2, 3, \dots, T-1\}$ we have*

$$\mathcal{V}_{T-j}(n, \pi) = \mathcal{E}(\pi) + (\bar{u} - \underline{u}) \mathcal{W}_{T-j}(n), \quad (\text{A.7})$$

where

$$\mathcal{W}_{T-j}(n) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}_{T-j+1}(n-y)) \right) \text{ and } \mathcal{W}_T(n) = 0 \quad (\text{A.8})$$

Furthermore, the reservation price for selling $1 \leq i \leq n$ units at $t = T-j$ equals

$$\mathcal{R}_{i,n,j} = \Delta \mathcal{W}_{T-j}(n-i+1) = \mathcal{W}_{T-j}(n-i+1) - \mathcal{W}_{T-j}(n-i) \quad (\text{A.9})$$

which weakly decreases as inventory increases.

Proof of Proposition A.2

We proceed by induction for general utility payoffs $\bar{u} > \underline{u}$ of the lottery, as well as probability π . At $t = T$ continuation value is $\mathcal{V}_T(n, \pi) = \mathcal{E}(\pi)$. Then, when $j = 1$ value in (A.6) is

$$\mathcal{V}_{T-1}(n, \pi) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} \mathcal{E}(\pi + Py) \right) + (1 - \beta) \mathcal{E}(\pi).$$

Since $\mathcal{E}(\pi + py) = \mathcal{E}(\pi) + py(\bar{u} - \underline{u})$ we obtain

$$\mathcal{V}_{T-1}(n, \pi) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} (\mathcal{E}(\pi) + Py(\bar{u} - \underline{u})) \right) + (1 - \beta) \mathcal{E}(\pi).$$

As in the last stage, $t = T$, it is optimal to sell as many units as demand permits, we deduce that

$$\mathcal{V}_{T-1}(n, \pi) = \mathcal{E}(\pi) + (\bar{u} - \underline{u}) \beta \mathbb{E}(P) \mathbb{E}(\min(n, Q)).$$

Next, we let $\mathcal{W}_{T-1}(n) \equiv \beta \mathbb{E}(P) \mathbb{E}(\min(n, Q))$ and thus

$$\mathcal{V}_{T-1}(n, \pi) = \mathcal{E}(\pi) + (\bar{u} - \underline{u}) \mathcal{W}_{T-1}(n)$$

Next, when $j = 2$ we obtain that (A.6) is

$$\mathcal{V}_{T-2}(n, \pi) = \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} \mathcal{V}_{T-1}(n - y, \pi + Py) \right) + (1 - \beta) \mathcal{E}(\pi).$$

Plugging $\mathcal{V}_{T-1}(n, \pi)$ and using $\mathcal{E}(\pi + py) = \mathcal{E}(\pi) + py(\bar{u} - \underline{u})$ we deduce

$$\begin{aligned} \mathcal{V}_{T-2}(n, \pi) &= \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} (\mathcal{E}(\pi + Py) + (\bar{u} - \underline{u}) \mathcal{W}_{T-1}(n - y)) \right) + (1 - \beta) \mathcal{E}(\pi) \\ \mathcal{V}_{T-2}(n, \pi) &= \mathcal{E}(\pi) + (\bar{u} - \underline{u}) \beta \mathbb{E} \left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}_{T-1}(n - y)) \right). \end{aligned}$$

We define the last term as in (A.8) and obtain

$$\mathcal{V}_{T-2}(n, \pi) = \mathcal{E}(\pi) + (\bar{u} - \underline{u}) \mathcal{W}_{T-2}(n).$$

Finally, for any $j \in \{3, 4, \dots, T-1\}$ we use induction. If $\mathcal{V}_{T-j+1}(n, \pi)$ obeys (A.7) and $\mathcal{W}_{T-j+1}(n)$ obeys (A.8) then

$$\mathcal{V}_{T-j}(n, \pi) = \mathcal{E}(\pi) + (\bar{u} - \underline{u})\beta\mathbb{E}\left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}_{T-j+1}(n - y))\right) = \mathcal{E}(\pi) + (\bar{u} - \underline{u})\mathcal{W}_{T-j}(n).$$

Next, to show that reservation values obey (A.9) and weakly decrease as inventory increases, we first show that $\mathcal{W}_{T-j}(n)$ is a concave function of n for any $j \in \{1, 2, \dots, T-1\}$. We proceed by induction, using (A.8). For $j = 1$, since $\mathcal{W}_T(n) = 0$ we obtain

$$\mathcal{W}_{T-1}(n) = \beta\mathbb{E}\left(\max_{y \in [0, \min(n, Q)]} Py\right) = \beta\mathbb{E}(P \min(n, Q))$$

This means that $\mathcal{W}_{T-1}(n)$ is a concave function of n . For $\min(n, q)$ is a concave function, and so is its expectation. A direct implication of concavity is that reservation prices are value-first differences. That is, the reservation price for selling i units at $t = T-1$ is

$$\mathcal{R}_{i,n,1} = \Delta\mathcal{W}_{T-1}(n - i + 1) = \mathcal{W}_{T-1}(n - i + 1) - \mathcal{W}_{T-1}(n - i)$$

Next, when $j = 2$ we obtain

$$\mathcal{W}_{T-2}(n) = \beta\mathbb{E}\left(\max_{y \in [0, \min(n, Q)]} (Py + \mathcal{W}_{T-1}(n - y))\right)$$

As in the proof of Theorem 1 in Carrasco and Smith (2017), we now rewrite the right-hand side maximization as a minimization problem where the choice variable is the post trade inventory level $z = n - y$. That is,

$$\mathcal{W}_{T-2}(n) = \beta\mathbb{E}\left(\max_{n - \min(n, Q) \leq z \leq n} (P(n - z) + \mathcal{W}_{T-1}(z))\right) = \beta\mathbb{E}(P)n - \beta\mathbb{E}\left(\min_{\max(n - Q, 0) \leq z \leq n} (Pz - \mathcal{W}_{T-1}(z))\right)$$

Finally, we eliminate the constraint by using the characteristic function $\chi_{C(q)}(z, n) = 0$ if $(z, n) \in C(q)$ and $+\infty$ otherwise, where $C(q) = \cup_n \{(z, n) | \max(n - q, 0) \leq z \leq n\}$. Then:

$$\mathcal{W}_{T-2}(n) = \beta\mathbb{E}(P)n - \beta\mathbb{E}\left(\min_{z \geq 0} (Pz - \mathcal{W}_{T-1}(z) + \chi_{C(Q)}(z, n))\right)$$

The first term $\beta\mathbb{E}(P)n$ is linear in n and so the concavity of the second term suffices for the concavity of $\mathcal{W}_{T-2}(n)$. As $C(q)$ is convex, then $\chi_{C(q)}(z, n)$ is convex in (z, n) . Furthermore, $-\mathcal{W}_{T-1}(z)$ is convex in z and so $Pz - \mathcal{W}_{T-1}(z) + \chi_{C(q)}(z, n)$ is convex in (z, n) . That is, according to Theorem 5.3 of Rockafellar (1970), $\min_{z \geq 0}(Pz - \mathcal{W}_{T-1}(z) + \chi_{C(q)}(z, n))$ is convex in n . The expectation preserves convexity and so $W_{T-2}(n)$ is concave in n . Using exactly the same inductive logic, it follows that $W_{T-j}(n)$ is concave for the remaining values $j \in \{3, 4, \dots, T-1\}$. Furthermore, the concave value means that the reservation prices for selling $1 \leq i \leq n$ units at $t = T - j$ obeys A.9 and weakly decrease as inventory increases.

Corollary 2 (*Single unit case and no demand restrictions*) When $n = 1$, the continuation value function for $j \in \{1, 2, 3, \dots, T-1\}$ solves

$$\mathcal{V}_{T-j} = \mathcal{E}(\pi) + (\bar{u} - \underline{u})\phi_{T-j}, \quad (\text{A.10})$$

where

$$\phi_{T-j} = \beta\mathbb{E}(\max(P, \phi_{T-j+1})) \text{ and } \phi_T = 0. \quad (\text{A.11})$$

Proof of Corollary 2

When $n = 1$ then (A.8) becomes (A.11) and (A.7) becomes (A.10) which completes the proof.

Corollary 3 (*Sell-all price*) At stage $t = T - j$ the sell-all reservation price is $\mathcal{R}_{n,n,j} = \mathcal{W}_{T-j}(1)$ for all $j \in \{1, 2, \dots, T-1\}$ and obeys

$$\mathcal{R}_{n,n,j} = \beta\mathbb{E}(\max(P, \mathcal{R}_{n,n,j-1})) \text{ and } \mathcal{R}_{n,n,0} = 0 \quad (\text{A.12})$$

Hence, it is invariant to inventory adjustments.

Proof: This follows immediately from A.9. The sell-all price is the price at which the seller is willing to sell all n units of inventory that he has; hence, $\mathcal{R}_{n,n,j} = \mathcal{W}_{T-j}(1)$. Exploiting A.8, we deduce that reservation prices obey A.12 and that they are invariant to inventory adjustments.

Corollary 4 (*Reservation price without demand restrictions*) When the seller faces no demand restrictions (i.e., $n \leq q$ for all values of q) then the seller uses a common reservation price $\mathcal{R}_j$ for each $j \in \{1, 2, \dots, T-1\}$ to either sell all or nothing. In this case, reservation prices obey A.12.

Proof: We proceed by induction. Let $\mathcal{R}_0 = 0$, so $\mathcal{W}_T(n) = n\mathcal{R}_0 = 0$. Suppose that $\mathcal{W}_{T-j+1}(n) = n\mathcal{R}_{j-1}$. When demand is unrestricted, A.8 gives

$$\begin{aligned}\mathcal{W}_{T-j}(n) &= \beta \mathbb{E} \left[\max_{y \in \{0, \dots, n\}} \{Py + (n-y)\mathcal{R}_{j-1}\} \right] \\ &= n\beta \mathbb{E} [\max\{P, \mathcal{R}_{j-1}\}] \\ &= n\mathcal{R}_j,\end{aligned}$$

where $\mathcal{R}_j = \beta \mathbb{E}[\max\{P, \mathcal{R}_{j-1}\}]$. Thus, $\mathcal{W}_{T-j}(n) = n\mathcal{R}_j$ for every $j \in \{1, \dots, T-1\}$. Consequently, $\mathcal{W}_{T-j}(n-i+1) - \mathcal{W}_{T-j}(n-i) = \mathcal{R}_j$ so all reservation prices equal $\mathcal{R}_j$ and satisfy the recursion in A.12. It is therefore optimal to sell all units when the price exceeds $\mathcal{R}_j$ and none when it is below $\mathcal{R}_j$; at equality, all feasible quantities yield the same payoff.

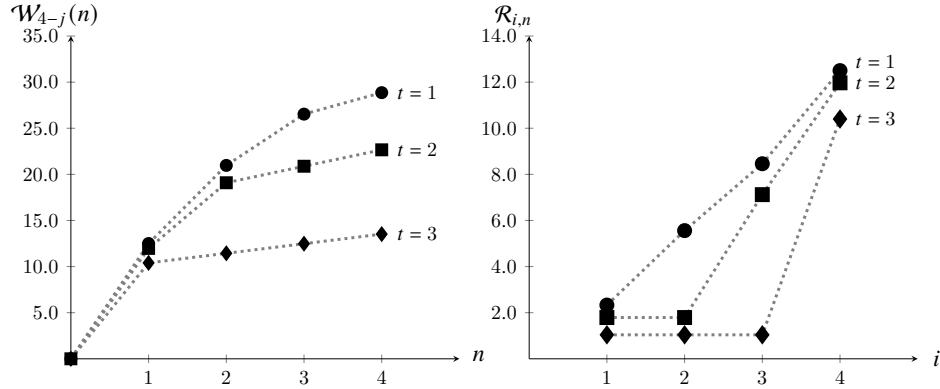
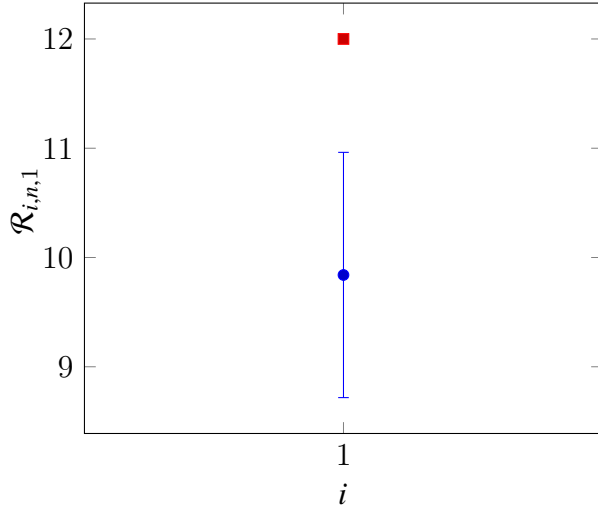


Figure A.1: **Values and reservation prices under a deadline $T = 4$.** Demand is 1 or 4 with probabilities $\alpha_1 = 0.9$ and $\alpha_4 = 0.1$, $\beta = 0.8$, and prices are uniformly distributed on $[1, 25]$. The left panel shows the declining continuation value as the deadline approaches. The right panel shows reservation prices for $n = 4$, which also decline over time. In period $t = 4 - j$, for $n = 4 - k$, $\mathcal{R}_i, n - k, j = \mathcal{R}_i + k, n, j$ for all $i \leq 4 - k$.

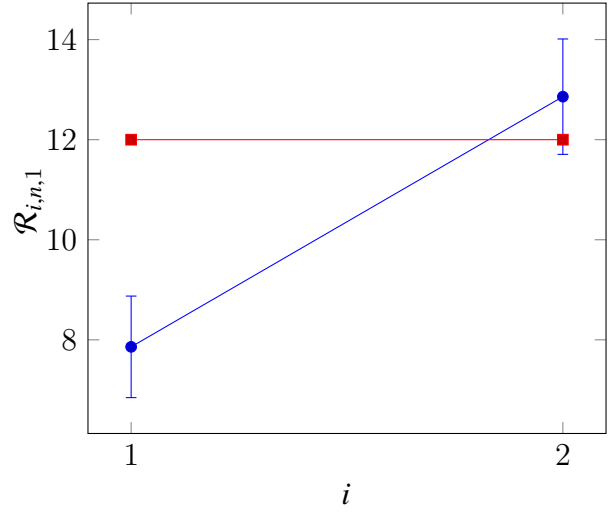
Config.	Unit	No Deadline		Deadline	
		Theoretical	Empirical	Theoretical	Empirical
A	1	13	10.22 (0.60)	12	9.84 (0.56)
B	1	13	8.27 (0.39)	12	7.86 (0.51)
	2	13	13.09 (0.57)	12	12.86 (0.58)
C	1	10	8.65 (0.44)	8	7.86 (0.37)
	2	13	12.77 (0.56)	12	12.27 (0.50)
D	1	9	10.30 (0.55)	7	9.74 (0.56)
E	1	13	6.43 (0.38)	12	6.18 (0.36)
	2	13	9.37 (0.46)	12	9.12 (0.45)
	3	13	12.19 (0.58)	12	11.90 (0.59)
	4	13	15.67 (0.66)	12	16.08 (0.70)
F	1	7	8.22 (0.44)	2	7.12 (0.41)
	2	8	11.18 (0.49)	2	10.53 (0.46)
	3	10	13.96 (0.56)	8	13.33 (0.55)
	4	13	17.18 (0.65)	12	16.48 (0.66)
G	1	9	8.17 (0.45)	7	7.47 (0.45)
	2	9	12.45 (0.54)	7	12.63 (0.59)
H	1	6	8.34 (0.47)	1	8.28 (0.44)
	2	8	12.36 (0.56)	2	12.64 (0.56)

Table A.1: Reservation values by unit and trading state across configurations, with and without deadlines. Reservation prices are reported in lottery-ticket units.

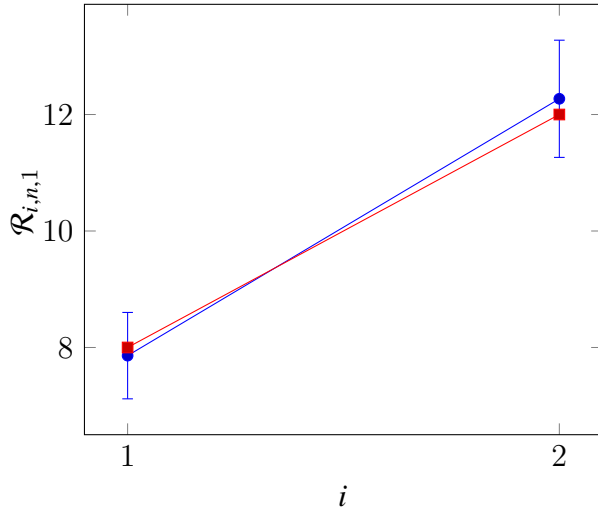
B Supply curves with deadline



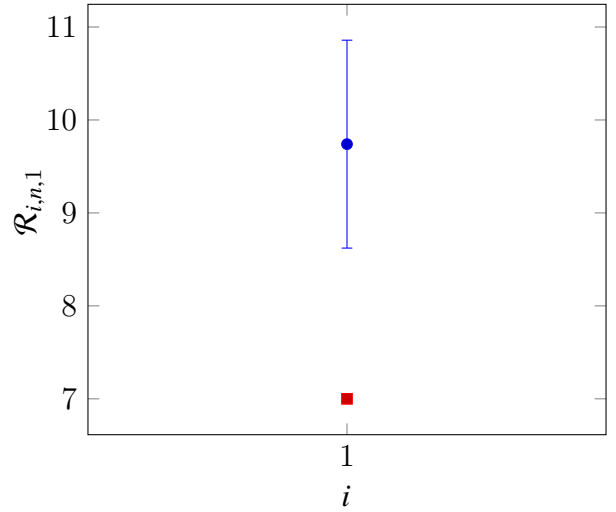
(a) Configuration A. Inventory: one unit. Demand distribution: $\alpha_1 = 1$.



(b) Configuration B. Inventory: two units. Demand distribution: $\alpha_2 = 1$.

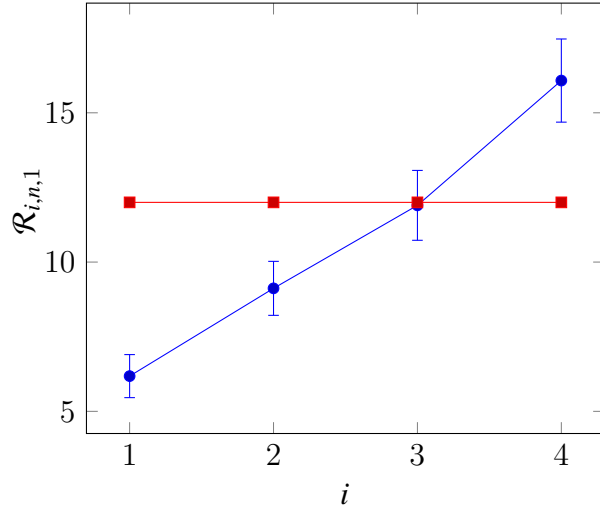


(c) Configuration C. Inventory: two units. Demand distribution: $\alpha_1 = 0.9$ and $\alpha_2 = 0.1$.

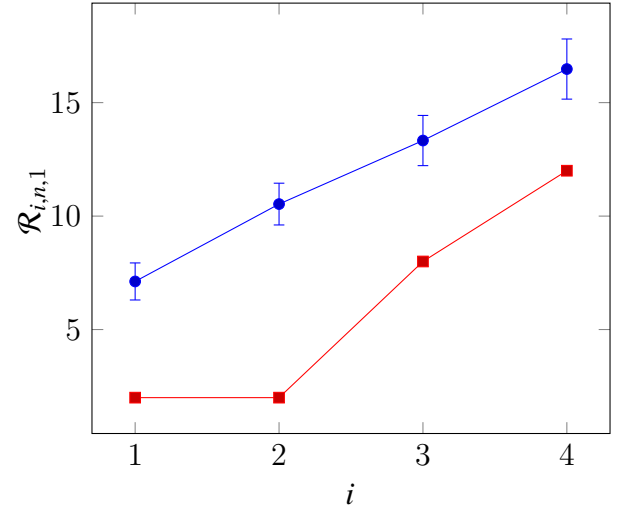


(d) Configuration D. Inventory: two units. Demand distribution: $\alpha_1 = 1$.

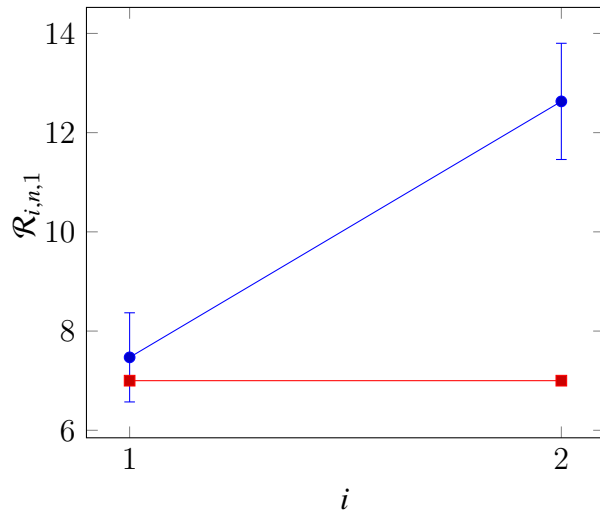
Figure B.1: Comparison of Configurations A, B, C, and D with a deadline $T = 3$. Theoretically predicted supply curves are shown in red, while empirical supply curves with confidence intervals are shown in blue. The empirical supply curves reflect average reservation prices for each unit in the first stage of the search process. Error bars indicate 95% confidence intervals, computed using clustered standard errors from a regression of reservation prices on unit dummies.



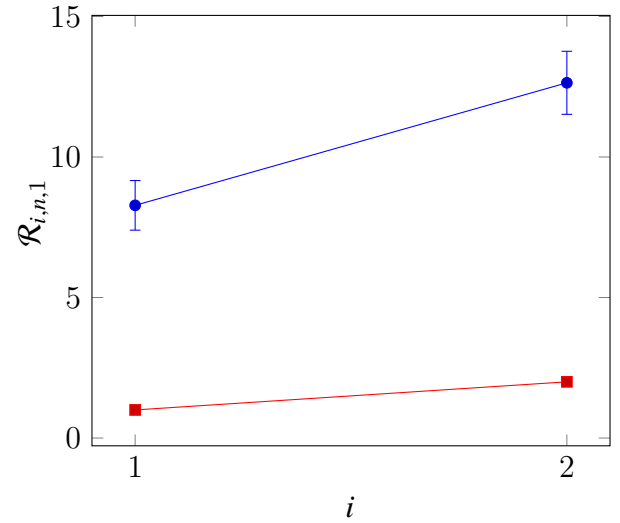
(a) Configuration E. Inventory: four units. Demand distribution: $\alpha_4 = 1$.



(b) Configuration F. Inventory: four units. Demand distribution: $\alpha_1 = 0.9$ and $\alpha_4 = 0.1$.



(c) Configuration G. Inventory: four units. Demand distribution: $\alpha_2 = 1$.



(d) Configuration H. Inventory: four units. Demand distribution: $\alpha_1 = 0.9$ and $\alpha_2 = 0.1$.

Figure B.2: Comparison of Configurations E, F, G, and H with a deadline $T = 3$. Theoretically predicted supply curves are shown in red, while empirical supply curves with confidence intervals are shown in blue. The empirical supply curves reflect average reservation prices for each unit in the first stage of the search process. Error bars indicate 95% confidence intervals, computed using clustered standard errors from a regression of reservation prices on unit dummies.

C Additional Empirical Results

	Mean	Std. Dev.
Male	0.46	0.50
Age	38.03	12.15
English	0.96	0.20
Student	0.13	0.34
Employed Full Time	0.41	0.49
Employed Part Time	0.17	0.38
CRT Score	1.66	1.23
Subjects	201.00	

Notes: Male, English, Student, Employed Full Time, and Employed Part Time are equal to one if the subjects report being male, English as their primary language, being a student, being employed full time, or being employed part time, respectively. CRT Score is the number of correct answers on a Cognitive Reflection Test, ranging from 0 to 3.

Table C.1: Summary Statistics

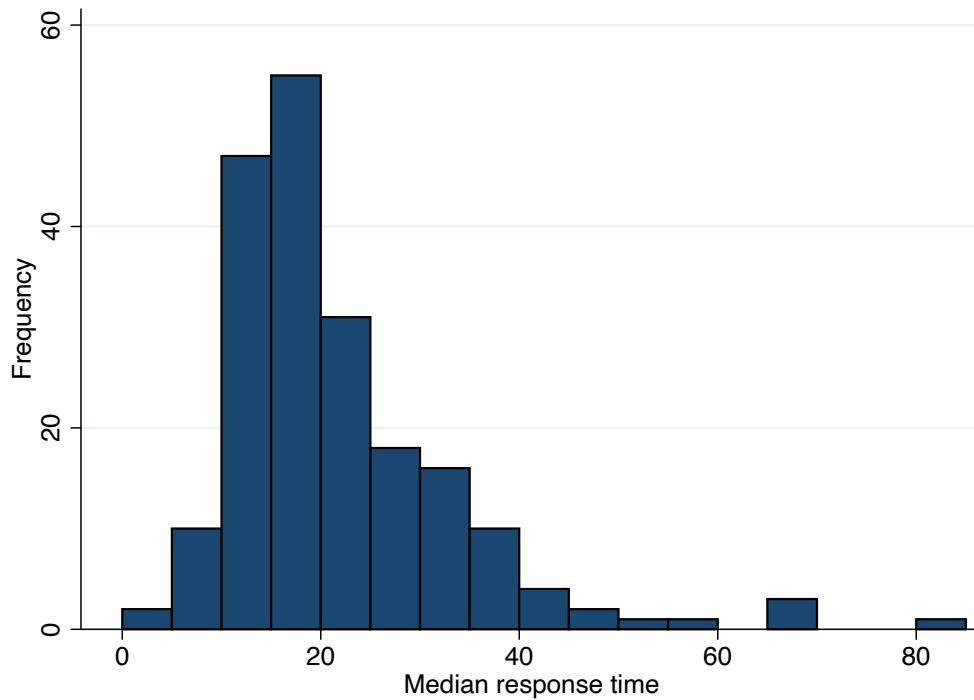


Figure C.1: Median Response Times

	(1)	(2)
	Response Time	Response Time
Inventory	2.89*	2.31
	(1.69)	(1.63)
Average Demand	-1.48	-2.49*
	(1.56)	(1.42)
Max Demand	6.84***	7.82***
	(2.09)	(1.98)
Search Number		-5.93***
		(0.47)
Constant	10.5***	38.4***
	(2.51)	(2.44)
Observations	1608	1608

Notes: Linear regression with subject fixed effects and standard errors clustered at the subject level. Significance indicated by:
*** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Response Times

	(1)	(2)	(3)	(4)
	Res. Price	Res. Price	Res. Price	Res. Price
Higher Inventory	-0.28 (0.28)		0.014 (0.23)	
Higher Inventory $\times$ Predicted Effect		0.022 (0.067)		-0.071** (0.035)
Constant	10.4*** (0.32)	10.3*** (0.32)	9.78*** (0.33)	9.61*** (0.32)
Type	No Deadline	No Deadline	Deadline	Deadline
Observations	1010	1010	1000	1000

Notes: Linear regression with subject and demand fixed effects and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.3: Effect of inventory levels on reservation prices (demand fixed effects)

	(1)	(2)	(3)	(4)
	Deviation	Deviation	Abs(Deviation)	Abs(Deviation)
Inventory	2.02*** (0.094)	2.02*** (0.094)	0.38*** (0.11)	0.38*** (0.11)
Unit Number	1.57*** (0.13)	1.57*** (0.13)	0.45*** (0.088)	0.44*** (0.087)
Average Demand	-2.67*** (0.099)	-2.66*** (0.10)	-0.17* (0.099)	-0.16 (0.099)
Search Number		0.031 (0.051)		0.098*** (0.036)
Constant	-3.04*** (0.36)	-3.18*** (0.42)	3.60*** (0.26)	3.14*** (0.32)
Observations	3618	3618	3618	3618

Notes: Linear regression with subject fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

Table C.4: Deviations from the theoretical benchmark

	(1)	(2)
	Res. Price	Res. Price
Unit # $\times$ Config. B	4.82*** (0.35)	5.00*** (0.41)
Unit # $\times$ Config. C	4.12*** (0.30)	4.41*** (0.31)
Unit # $\times$ Config. E	3.06*** (0.19)	3.25*** (0.22)
Unit # $\times$ Config. F	2.97*** (0.19)	3.09*** (0.19)
Constant	4.16*** (0.39)	3.33*** (0.42)
Type	No Deadline	Deadline
Observations	1212	1200

Notes: Linear regression with Subject $\times$ Configuration fixed effects and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.5: Supply Curve Slopes with Unrestricted Demand

	(1)	(2)	(3)	(4)
	Res. Price	Res. Price	Res. Price	Res. Price
Higher Inventory $\times$ Predicted Effect	-0.063 (0.10)	0.15* (0.089)	-0.067 (0.044)	0.030 (0.053)
Constant	9.99*** (0.53)	10.7*** (0.39)	9.08*** (0.45)	10.3*** (0.45)
Type	No Deadline	No Deadline	Deadline	Deadline
CRT	Low	High	Low	High
Observations	400	610	450	550

Notes: Linear regression with subject and demand-unit fixed effects and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.6: Effect of inventory levels on reservation prices by CRT

	(1)	(2)
	Res. Price	Res. Price
Deadline $\times$ Predicted Effect	0.16 (0.12)	-0.079 (0.10)
Constant	9.59*** (0.61)	10.4*** (0.56)
Configuration-unit FE	Yes	Yes
CRT	Low	High
Observations	1530	2088

Notes: Linear regression with subject random effects, configuration-unit fixed effects, and standard errors clustered at the subject level. Significance indicated by: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.7: The effect of deadlines on reservation prices by CRT

	(1)	(2)	(3)	(4)
	Deviation	Deviation	Abs(Deviation)	Abs(Deviation)
Inventory	2.28*** (0.14)	1.83*** (0.12)	0.47*** (0.16)	0.32** (0.15)
Unit Number	1.92*** (0.20)	1.31*** (0.16)	0.57*** (0.14)	0.35*** (0.11)
Average Demand	-2.73*** (0.16)	-2.61*** (0.13)	-0.26* (0.15)	-0.083 (0.13)
Search Number	0.033 (0.075)	0.038 (0.070)	0.12** (0.058)	0.084* (0.047)
Constant	-4.24*** (0.67)	-2.45*** (0.52)	2.79*** (0.56)	3.37*** (0.38)
CRT	Low	High	Low	High
Observations	1530	2088	1530	2088

Notes: Linear regression with subject fixed effects and standard errors clustered at the subject level. Significance indicated by: *** p<0.01, ** p<0.05, * p<0.1.

Table C.8: Deviations from theoretical benchmark by CRT

Online Appendix

D Screenshots/instructions

Below, we include screenshots from the experiment.

Instructions

Thank you for participating in the study! This study is about how people make decisions. Specifically, we will study how people choose to sell items when they are uncertain about their opportunities to sell items in the future. On the bottom of every page of the study, you will be given a short summary of the rules we show on this page for reference.

The study will be split into **8 rounds**, and each round will be split into **trading stages**.

You will start each **round** with a stock of (imaginary) lobsters that you can sell, but that have no value to you if you don't sell them. You will be able to trade these lobsters for lottery tickets. At the end of the study, we will randomly select one round to be the one that counts. Each lottery ticket that you earned in that round will be worth a 1% chance of receiving a **bonus payment of \$10**.

In each **trading stage**, you will be matched with a computerized buyer who wants to trade lottery tickets for lobsters. The **per-unit price** (in lottery tickets) they are willing to pay and the **demand** (i.e. number of lobsters that the buyer wants) will be randomly drawn in each trading stage. The buyer's ticket offer will always be between 1 and 25 tickets per lobster, with each value being equally likely to be drawn. However, the chances of each amount of demand will change from round to round (but not between stages within rounds).

Before you see the buyer's offer in a trading stage, you will decide the **minimum number of tickets you are willing to trade for each of your lobsters**. For instance, you could state that you would be willing to sell one lobster for 5 lottery tickets, but would only sell two lobsters for a higher price of 7 tickets per lobster. After you make your choices, your minimum acceptable prices will be compared to the buyer's random demand and ticket offer. You will sell either the number demanded by the buyer or the maximum number you were willing to sell (whichever is lower) at the price in lottery tickets offered by the buyer.

After each stage, there is a 20% chance that all of your remaining lobsters (those that have not already been sold) will spoil, and you will not be able to sell them anymore. To determine whether that will happen, the computer will roll a 100-sided die. If the result is higher than 80, then the lobsters spoil and the round ends. If the result is less than or equal to 80, then you move on to the next stage. Even if your lobsters spoil, you keep any lottery tickets you have already received.

Once you have sold all of your lobsters or your remaining lobsters spoil, then you move on to the next round. If all rounds have been completed, then you will finish the study and find out whether you received the bonus.

On the next few pages, you will see examples of the types of decisions you will face. Look through them to make sure you understand how the study will work.

Next

Figure D.1: Instructions - No Deadline

Instructions

Thank you for participating in the study! This study is about how people make decisions. Specifically, we will study how people choose to sell items when they are uncertain about their opportunities to sell items in the future. On the bottom of every page of the study, you will be given a short summary of the rules we show on this page for reference.

The study will be split into **8 rounds**, and each round will be split into **trading stages**.

You will start each **round** with a stock of (imaginary) lobsters that you can sell, but that have no value to you if you don't sell them. You will be able to trade these lobsters for lottery tickets. At the end of the study, we will randomly select one round to be the one that counts. Each lottery ticket that you earned in that round will be worth a 1% chance of receiving a **bonus payment of \$10**.

In each **trading stage**, you will be matched with a computerized buyer who wants to trade lottery tickets for lobsters. The **per-unit price** (in lottery tickets) they are willing to pay and the **demand** (i.e. the maximum number of lobsters that the buyer wants) will be randomly drawn in each trading stage. The buyer's ticket offer will always be between 1 and 25 tickets per lobster, with each value being equally likely to be drawn. However, the chances of each amount of demand will change from round to round (but not between stages within rounds).

Before you see the buyer's offer in a trading stage, you will decide the **minimum number of tickets you are willing to trade for each of your lobsters**. For instance, you could state that you would be willing to sell one lobster for 5 lottery tickets, but would only sell two lobsters for a higher price of 7 tickets per lobster. After you make your choices, your minimum acceptable prices will be compared to the buyer's random demand and ticket offer. You will sell either the number demanded by the buyer or the maximum number you were willing to sell (whichever is lower) at the price in lottery tickets offered by the buyer.

After each stage, there is a 20% chance that all of your remaining lobsters (those that have not already been sold) will spoil, and you will not be able to sell them anymore. To determine whether that will happen, the computer will roll a 100-sided die. If the result is higher than 80, then the lobsters spoil and the round ends. If the result is less than or equal to 80, then you move on to the next stage. There is also a limit to the number of stages in each round - once that limit is reached, all of your remaining lobsters are guaranteed to spoil and the round ends. Even if your lobsters spoil, you keep any lottery tickets you have already received.

Once you have sold all of your lobsters or your remaining lobsters spoil, then you move on to the next round. If all rounds have been completed, then you will finish the study and find out whether you received the bonus.

On the next few pages, you will see examples of the types of decisions you will face. Look through them to make sure you understand how the study will work.

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Figure D.2: Instructions - Deadline

Example 1: Alice

In Round 1, Alice starts with 1 lobster. She knows that in each stage, a buyer will arrive offering to buy 1 lobster.

Stage 1: Alice starts the stage with 1 lobster and 0 tickets. She states that she will sell her lobster if the price offer is at least 8 tickets. The buyer offers to buy one lobster for 3 tickets. Because the offer is lower than Alice's minimum price, Alice does not sell any lobsters and the stage ends. The computer rolls a 100-sided die, and the result is less than 80, so the round continues.

Stage 2: Alice starts the stage with 1 lobster and 0 tickets. She states that she will sell her lobster if the price offer is at least 6 tickets. The buyer offers to buy one lobster for 11 tickets. Because the offer is higher than Alice's minimum price, Alice sells the lobster and receives the 11 lottery tickets. Because Alice sold all of her lobsters, the round is over.

Alice earned 11 tickets in total this round, so if this is the round that counts she has an 11% chance of receiving \$10.

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Figure D.3: Example 1

Example 2: Bob

In Round 2, Bob starts with 2 lobsters. He knows that in each stage, a buyer will arrive offering to buy 1 lobster.

Stage 1: Bob starts the stage with 2 lobsters and 0 tickets. He states that he will sell 1 lobster if the price offer is at least 9 tickets. The buyer offers to buy 1 lobster for 12 tickets. Because the offer is higher than Bob's minimum price, Bob sells 1 lobster and earns 12 tickets. The computer rolls a 100-sided die, and the result is less than 80, so the round continues.

Stage 2: Bob starts the stage with 1 lobster and 12 tickets. He states that he will sell his lobster if the price offer is at least 6 tickets. The buyer offers to buy 1 lobster for 3 tickets. Because the offer is lower than Bob's minimum price, Bob does not sell any lobster and the stage ends. The computer rolls a 100-sided die, and the result is less than 80, so the round continues.

Stage 3: Bob starts the stage with 1 lobster and 12 tickets. He states that he will sell his lobster if the price offer is at least 7 tickets. The buyer offers to buy 1 lobster for 1 ticket. Because the offer is lower than Bob's minimum price, Bob does not sell any lobsters and the stage ends. The computer rolls a 100-sided die, and the result is higher than 80, so the lobsters spoil and the round ends.

Bob earned 12 tickets in total this round, so if this is the round that counts he has a 12% chance of receiving \$10.

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Figure D.4: Example 2

Example 3: Charlie

In Round 3, Charlie starts with 4 lobsters. He knows that in each stage, there is a 50% chance that the buyer will offer to buy 1 lobster, and a 50% chance that the buyer offers to buy up to 2 lobsters.

Stage 1: Charlie starts the stage with 4 lobsters and 0 tickets. He states that he will sell 1 lobster if the price offer is at least 5 tickets and 2 lobsters if the price offer is at least 9 tickets. The buyer offers to buy up to 2 lobsters for 15 tickets each. Because the offer is higher than Charlie's minimum price for 2 lobsters, Charlie sells 2 lobsters and earns 30 tickets. The computer rolls a 100-sided die, and the result is less than 80, so the round continues.

Stage 2: Charlie starts the stage with 2 lobsters and 30 tickets. He states that he will sell 1 lobster if the price offer is at least 4 tickets and 2 lobsters if the price offer is at least 11 tickets. The buyer offers to buy up to 2 lobsters for 9 tickets each. Because the offer is higher than Charlie's minimum price for 1 lobster but lower than his minimum price for 2 lobsters, Charlie sells 1 lobster and earns 9 tickets. The computer rolls a 100-sided die, and the result is higher than 80, so the lobsters spoil and the round ends.

Charlie earned 39 tickets in total this round, so if this is the round that counts he has a 39% chance of receiving \$10.

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Figure D.5: Example 3

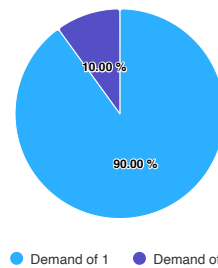
Round 1, Stage 1

You currently have a stock of **4 lobsters** and so far, you have 0 lottery tickets in this round.

There is a 20% chance that your lobsters will spoil after this stage.

In this round, there is a 90% chance that the number of lobsters demanded is 1 and 10% chance that it is 2. That means that given your stock, the maximum you could sell is 2.

Demand Chances



In the boxes below, please list the minimum number of lottery tickets per lobster you would be willing to accept to sell each number of lobsters up to 2. Remember that the numbers you list must be between 1 and 25, and the minimum number of tickets must increase with the number of lobsters.

Minimum price per unit to sell 1 unit

Minimum price per unit to sell 2 units

Next

Figure D.6: Reservation Prices - No Deadline

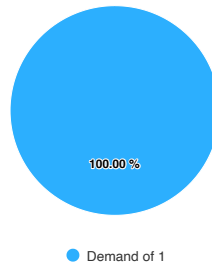
Round 1, Stage 1

You currently have a stock of **1 lobster** and so far, you have 0 lottery tickets in this round.

There is a 20% chance that your lobsters will spoil after this stage. Furthermore, there are **3 stages left**, including this one, before all of your lobsters are guaranteed to spoil.

In this round, the number of lobsters demanded is guaranteed to be 1. That means that given your stock, the maximum you could sell is 1.

Demand Chances



In the box below, please list the minimum number of lottery tickets you would be willing accept to sell your lobster. Remember that the price you list must be between 1 and 25.

Minimum price per unit to sell 1 unit

Next

Figure D.7: Reservation Prices - Deadline

Results: Round 1, Stage 1

In this stage, the buyer offered 21 lottery tickets per lobster. On the previous page, you reported that the minimum you would accept to sell one lobster is 25. Because the random price is below this, you have not agreed to sell any lobsters at this price.

This stage's random demand was 1. That means that you sold 0 lobsters and received 0 lottery tickets this stage.

You have accumulated a total of 0 lottery tickets through sales in this round.

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Figure D.8: Stage Results - No Sales

Results: Round 2, Stage 1

In this stage, the buyer offered 22 lottery tickets per lobster. That is higher than the highest minimum acceptable price you reported on the previous page, so you have agreed to sell up to 4 lobsters at this price.

This stage's random demand was 4. That means that you sold 4 lobsters and received 88 lottery tickets this stage.

You have accumulated a total of 88 lottery tickets through sales in this round.

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Figure D.9: Stage Results - Sales

Continuation: Round 1, Stage 1

You sold your full stock so this round is over. You accumulated a total of 21 lottery tickets in this round, so if this round is chosen to be the one that counts, you have a 21% of receiving a bonus payment of \$10.

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Figure D.10: Continuation - Full inventory Sold

Continuation: Round 4, Stage 1

The computer randomly rolled a 96, which is higher than 80. That means that this round is over.

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Figure D.11: Continuation - Random Termination

Continuation: Round 3, Stage 3

All of your lobsters were guaranteed to spoil after the last stage, so this round is over. You accumulated a total of 0 lottery tickets in this round, so if this round is chosen to be the one that counts, you have a 0% of receiving a bonus payment of \$10.

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Figure D.12: Continuation - Deadline Reached

Final Results

That was the last round. The round that counts was number 8. In that round, you accumulated a 29.0% chance of winning the prize.

We can now inform you that **you did not win the prize**. That means you will not receive a bonus.

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Figure D.13: Final Results